Lowering Barriers to Remote Education: Experimental Impacts on Parental Responses and Learning*

Emily A. Beam†
Priya Mukherjee‡
Laia Navarro-Sola§

October 22, 2024

Abstract

We conducted a randomized controlled trial in Bangladesh during the COVID-19 school closures to investigate how parents adjusted their educational investments in response to three interventions: information about an educational technology tool, an internet data package, and one-on-one phone support. None of the interventions directly improved student learning. However, they affected parental time and monetary investments in education, and some changes contributed to learning gains. Specifically, receiving the edtech information did not affect its uptake but increased tutoring

*This study was supported by the World Bank's Strategic Impact Evaluation Fund (SIEF) and the Stiftelsen Carl Mannerfelts fund and benefited in particular from support from Alaka Holla and Erin Kelley. We thank Junnatul (Ani) Ferdosh, Md. Afzal Hossain Sarwar, Avijit Saha, Rafiqul Islam Sujon, Sakhwat Hosain, and 71 a2i partner-teachers for their cooperation to execute the research. We thank Sathia Chakrapani and Sohini Chaparala for outstanding project management and support. We also thank Tanzila Tabassum, Sukriti Ahmed, Zaima Promy, Pavarin Bhandtivej, Ishmail Baako, and Samina Hossain for critical research support, as well as the team of enumerators and supervisors who diligently collected data in spite of the challenges of the lockdown. This paper has benefited greatly from the comments and suggestions of Tessa Bold, Sebastián Gallegos, Alejandro Ganimian, Seema Jayachandran, David Schönholzer, Abhijeet Singh, and seminar and conference participants at Stockholm University, ASWEDE conference, AEFP EdDev Workshop, Midwest International Economic Development Conference, Advances with Field Experiments conference, CEPR Annual Symposium in Labour Economics, Universidad Autonoma de Madrid, Universitat Autonoma de Barcelona, Universitat Pompeu Fabra, University of Essex, EIEF Rome, Lund University, University of Bergen, and the University of Richmond.

The Human Subjects Committee for Innovations for Poverty Action provided oversight for this project, "Bangladesh COVID-19 Remote Learning Technologies," protocol #15594. This study was pre-registered at the AEA RCT Registry as "Take-up, Use, and Effectiveness of Remote Learning Technologies," #AEARCTR-0006191.

†University of Vermont and IZA, emily.beam@uvm.edu
‡U. Wisconsin-Madison, priya.mukherjee@wisc.edu
§IIES, Stockholm University, laia.navarrosola@iies.su.se

investments, driving improvements in math achievement. The data package led to more modest increases in tutoring investments, while phone-based support reduced tutoring.

JEL Classification: C93, I21, I24, J13, O15
Keywords: Human capital, parental investments, educational technology, educational inequality

1 Introduction

Parental investments are an important determinant of children's skills and human capital (Becker and Tomes, 1976; Cunha et al., 2006; Todd and Wolpin, 2007; Francesconi and Heckman, 2016). Beyond selecting schools and formal learning environments, parents provide important supplemental educational inputs that complement formal schooling. These investments fall into two main categories: time investments—such as helping with homework or engaging in educational activities—and economic investments— such as paying for tutoring or other after-school activities (Bray, 1999). However, parents face barriers to optimizing these investments, including limited knowledge of educational options, low perceived returns to these investments (Attanasio et al., 2020), limited availability of different resources and their costs, and their own financial constraints (Dahl and Lochner, 2012). Addressing these barriers is particularly important for families with limited resources, who may benefit the most from educational investments, and in contexts of educational and schooling disruptions, such as during the COVID-19 school closures.

Understanding parental behavioral responses to these sorts of education policies is critical for accurately assessing their net impact on children's human capital development (Das et al., 2013) and anticipating potential distributional effects. Households often act as intermediaries between education policies and children's learning. Without accounting for these behavioral responses, it is challenging to determine, for example, whether an intervention was ineffective or if it succeeded but led parents to adjust their behavior, thereby offsetting intended effects. Additionally, inequality in parental time, skills, and financial resources can create disparities in educational investments in children, potentially worsening educational inequality (Blanden et al., 2022). Hence, examining how parental responses differ across households with varying socioeconomic backgrounds helps to understand the drivers of the distributional impacts of reducing barriers to education.

We conducted a randomized controlled trial with 7,576 households with secondary school students in Bangladesh to investigate how parents adjust their investments in response to three short-term interventions targeting different constraints on educational investments. The interventions were informational phone messages about a phone-based educational technology (edtech), an unconditional cash transfer delivered as an internet data package alongside the phone messages, and one-on-one phone teacher support. These treatments allowed us to experimentally assess the effects of addressing specific barriers to remote education: the information increased the salience of an edtech resource, the data package alleviated economic constraints related to internet access, and the teacher support provided personalized assistance to students. Our findings indicate that none of the treatments directly improved student learning. However, they did lead parents to adjust their educational investments, and some of these endogenous changes contributed to learning gains. Specifically, the information intervention led to increased investment in private tutoring, which appears to have improved student achievement.

We delivered interventions for 4–8 weeks from February to April 2021 to the phone number reached during the baseline survey, which was nearly always that of the child's parent. We conducted a follow-up survey by phone in March 2021 to measure the impact of these interventions on parent and student educational investments while the interventions were ongoing. We conducted a second follow-up survey in June 2021, approximately one month after the interventions concluded, to measure student learning.[1] First, we describe the nature of parental economic and financial investments during the COVID-19 school closures. Next, we investigate the impacts of the interventions on parental investments during the intervention period. Finally, we examine the impacts on student math achievement one to two months after the interventions ended.

That this study took place during the 2020–2021 COVID-19 school closures makes it particularly suitable for investigating the impacts of these three educational policies on parental responses and educational inequality. During periods of disrupted or limited access to quality schooling, parents become the primary decision-makers for their children's educational

[1]We pre-registered our primary empirical specification and key outcomes at https://www.socialscienceregistry.org/trials/6191.

inputs. This context, while different from traditional educational settings, facilitates the detection of parental behavioral responses and isolates their contribution to human capital development from other inputs like formal schooling. Differences in parents' ability and resources to support remote learning, compensate for lost school-based inputs, and respond to and benefit from interventions may deepen educational inequality (Fredriksson et al., 2016; Blanden et al., 2022; Agostinelli et al., 2022), with potential long-run implications (Fuchs-Schündeln et al., 2022).[2]

We first descriptively document parental investments in children's education during the COVID-19 school closures to understand how parents supported education when formal schooling was unavailable. We include households facing a broad range of constraints to investigate how socioeconomic status is associated with investments and responses to reduced barriers to remote education. Our findings show that nearly all students continued to pursue educational activities regularly one year after school closures, but relatively few households use tech-based resources to support learning, especially among poorer families. Parents reported an average of 6.6 hours per week helping their child with schoolwork, and 64% of households reported using private tutoring in the past month. Economic and time investments were positively correlated among wealthier households, while the economic investments of poorer households remained consistently lower and were not correlated with their time inputs, suggesting binding constraints to investment.

We then present the experimental findings showing that the three remote interventions were largely ineffective in improving student mathematics learning. However, these interventions did affect how parents allocated their time and monetary resources for education during the intervention period. Specifically, providing regular information about an edtech tool did not effectively encourage its use. Despite this, it led parents to adjust their investments, with some changes in the use of learning resources and significant increased economic

[2] List et al. (2021) show that in the United States, simple informational policies are not enough to change parental beliefs about the effectiveness of parental investments and that more intensive programs combining information and home visits and feedback are needed to increase parental investments and reduce socioeconomic gaps in children's achievement.

investment in private tutoring on the extensive and intensive margin. The unconditional cash transfer provided via a free data package to support internet use also increased private tutoring on the extensive margin and, only when combined with edtech information, did it lead to a greater edtech tool usage. Phone-based teacher support decreased the use of in-person teachers and reduced the likelihood of using a private tutor, particularly among poorer households.

These parental responses help explain the observed learning impacts. The edtech information treatment resulted in a 0.15 standard deviation increase in math learning. The lack of rise in edtech tool usage suggests that the increased tutoring investment likely drove these learning gains. The cash transfer through the data package led to more modest increases in tutoring investment, and we did not observe significant learning gains from this intervention. Finally, parents, particularly those with limited resources, temporarily reallocated tutoring investments while receiving phone teacher support, which may have offset potential learning gains from this treatment.

We contribute to three strands of literature. First, our results add to the literature on parental investments and involvement in their children's education. Previous research has investigated how exogenous changes in schooling inputs affect parental time investment at home, finding that it can substitute for school resources in countries like India and Zambia (Das et al., 2013) and Romania (Pop-Eleches and Urquiola, 2013). In contrast, studies in the United States have shown parental time investments to be either a complement (Gelber and Isen, 2013) or a substitute (Houtenville and Conway, 2008) for school resources. Our study experimentally examines parental educational investment responses to three common remote educational interventions in a context where school inputs have minimal influence on students' learning. By collecting detailed data on parental time and economic investments, as well as choices of learning resources, we show that households' investment responses vary depending on the specific intervention received—even though all interventions aim to reduce barriers to access—and that a joint re-optimization of educational inputs often results in shifts in investments above and beyond the targeted educational input targeted of the intervention. While the specific magnitudes of our estimates are context-specific, especially given the COVID-19 pandemic, the broader finding that these investment responses lead to sizeable and heterogeneous impacts on student outcomes is likely not unique to this setting.

Second, we contribute to the literature on interventions designed to improving educational outcomes during school disruptions caused by natural disasters and emergencies (Andrabi et al., 2021; Bandiera et al., 2020), including the COVID-19 pandemic. Research in this area has explored how school closures affect learning (Agostinelli et al., 2022) and create inequalities (Bacher-Hicks et al., 2021; Singh et al., 2022), as well as the experimental impacts of interventions aimed at maintaining student engagement and learning during such closures (Angrist et al., 2022; Carlana and La Ferrara, 2021; Lichand et al., 2022; Hassan et al., 2021; Schueler and Rodriguez-Segura, 2021). More broadly, we contribute to the literature on educational technology. Relatively low-tech solutions such as SMS and phone calls (Angrist et al., 2022), interactive voice-recorded lessons (Wang et al., 2023), and in-school TV-based lessons (Navarro-Sola, 2021; Johnston and Ksoll, 2017; Beg et al., 2019) have shown promise, as well as personalized adaptive computer-assisted learning (CAL) programs (Muralidharan et al., 2019) and personalized tutoring (Glewwe et al., 2024).[3] In contrast, our interventions did not directly promote learning. We identified several barriers to using the edtech tool, and even one-on-one teacher support was insufficient to improve learning outcomes, aligning with the null impacts found by Crawfurd et al. (2023).[4] Our findings underscore that merely providing resources is not enough to boost learning; effective interventions must consider the role of parents as intermediaries who may adjust their educational investments in response to the interventions.

Third, our paper contributes to the literature on how parental investments and constraints impact achievement gaps and educational inequality, extensively reviewed in Blan-

[3]See Caballero Montoya et al. (2021) for a thorough review of the literature on distance education.

[4]Our intervention also differs in duration from successful programs (4 weeks vs. 6 weeks in Angrist et al. (2022)) and teacher autonomy regarding the direction of remote lessons.

den et al. (2022). Households from lower socioeconomic backgrounds often face greater time and monetary constraints. Information frictions, which are greater for poorer families (Dizon-Ross, 2019), can exacerbate differences in parental investments in children's education (Caucutt et al., 2017). Schooling disruptions can increase the relative importance of parental investments, with higher-SES parents potentially better able to adjust their investments to mitigate these shocks (Blanden et al., 2022). However, most evidence to date is from wealthier contexts (Andrew et al., 2020; Del Bono et al., 2021; Bansak and Starr, 2021; Bacher-Hicks et al., 2021). Our study indicates that heterogeneous constraints among parents from different socioeconomic backgrounds mean that some policies designed to reduce educational barriers may, in fact, worsen educational inequality.

The paper is organized as follows. Section 2 describes the experimental design, including the context of education in Bangladesh during the COVID-19 school closures, sample selection, study timeline, description of the interventions, randomization, and attrition. It also presents descriptive statistics, balance tests, and the empirical specification. Section 3 provides descriptive insights into parental investments in children's education. Section 4 analyzes the impacts of the interventions on parent economic and time educational investments, as well as learning resource usage. Section 5 examines the persistent effects on student learning. Section 6 discusses potential mechanisms and channels behind the findings, and Section 7 concludes.

2 Experimental Design

2.1 Context: Education in Bangladesh during COVID-19

The first known cases of COVID-19 were reported in Bangladesh on March 7, 2020. Bangladesh initiated a general holiday on March 18, 2020, closing schools and all non-essential businesses. In October, the government issued assignments and evaluation guidelines for secondary-level students and announced that students would be automatically promoted to the next grade based on these evaluated assignments (Alamgir, 2020). In January 2021, the government announced plans to reopen schools in February, and it issued and distributed new books to students for the 2021 academic year.[5] The government reversed this decision as COVID-19 cases rose, and it did not re-open schools until September 2021, leaving Bangladesh with some of the longest school closures globally (UNESCO, 2022).[6]

During the school closures, the government's main priority was to minimize the disruption of learning as much as possible. The Ministry of Education and Aspire to Innovate (a2i), the government's tech arm, collaborated to use a mass media broadcasting, supplemented with an online platform, to remotely deliver educational content from the school curriculum. The government began broadcasting daily television lessons for secondary-level students on March 29, which was later expanded to all levels. The secondary broadcasts consisted of 10 videos daily—two grade-specific 20-minute daily lessons for students in grades 6 through 10—and these lessons were also posted on a YouTube channel. Weekly broadcast schedules were disseminated widely: schools asked teachers to share schedules with households and encouraged them to watch, and schedules were also posted online and broadcast over radio. However, Sangsad TV was broadcast via satellite, so non-subscribing households, as well as those without televisions, were not able to access materials. Additionally, the Sangsad TV channel stopped broadcasting secondary lessons in anticipation of an early 2021 school reopening, and so it only telecast lessons for grades 1 through 5 during the intervention period. The pool of videos posted on YouTube, however, remained available.

Non-governmental organizations also offered educational resources and initiatives to aid remote learning during school closures. One such resource was Robi 10-Minute School, a free website platform with an accompanying mobile application that provided free videos and adaptive quizzes aligned with national curriculum standards. More than 1.5 million students accessed its materials daily in 2020 (Axiata Group Berhad, 2020).

[5]The Bangladesh academic calendar follows the calendar year, beginning in January and ending in December.

[6]Appendix Figure A.1 outlines key events in Bangladesh affecting children's education alongside the study timeline.

Althougb schools remained closed for the duration of our study, from September 2020–June 2021, the rest of the economy was generally open. The nationwide "general holiday" ended by May 30, and all movement restrictions, which mainly closed shops after 8pm and restricted movement after 10pm, were lifted by September 1, 2020. There was a period of lockdowns in April 2021, which occurred after our first follow-up survey. While households may have continued to experience the lasting effects of these economic disruptions, members were not restricted from working during the time period of our study.

2.2 Sample Selection

Because the interventions are useful only to those who have access to the requisite technology, our baseline phone sample consists of 7,576 respondents that have (a) at least one child in grades 6–10 (grades 7–11 in January 2021) and (b) have at least one smartphone in the house, of which 7,313 agreed to be recontacted in follow-up waves. While mobile phone penetration in Bangladesh is fairly high, smartphone ownership is substantially lower, meaning that our study sample is not nationally representative of families with secondary school children. Estimates in 2022 put individual-level smartphone ownership at 41% (Okeleke, 2021), although rates of access are likely higher given that device sharing is common in Bangladesh (Ahmed et al., 2017).

Because we include only households with access to a smartphone, our sample is not nationally representative of families with secondary school children. We attempt to include a wide range of socioeconomic backgrounds despite this restriction by building our sample from three sources: (1) a random-digit-dialing (RDD) sample of 30,000 numbers from the most popular telecommunications company in Bangladesh; (2) the database of recipients of the Secondary School Stipend (SSS) Programme, who tend to be from lower-income households; and (3) a database of users registered on a government-created online learning platform that preceded the COVID-19 pandemic. While the RDD sample aims, by design, to be nationally representative of the smartphone ownership population, the secondary school stipend sample includes a higher share of lower-income households, and the last sample includes households potentially more inclined to use educational technologies during school closures. We first screened numbers by sending a test SMS message and removing any numbers for which the message was not delivered. Overall, 7,576 respondents completed a baseline survey, about 19% of numbers attempted, or 29% of those who answered the phone (see Appendix Table A.1 for more detail). Respondents are distributed broadly across the country (see Appendix Figure A.2). We randomized all baseline respondents into treatment, but we further restrict our sample to the 97% who agreed following baseline to be recontacted for follow-up surveys (7,313 households).

2.3 Interventions

We test the impact of three interventions designed to reduce different constraints to parental educational investment:

Treatment 1: Information about an educational technology (edtech) tool. Households received twice-weekly reminders about a free internet-based learning platform, Robi 10-Minute School, for eight weeks.[7] This resource had a webpage containing videos and adaptive quizzing aligned with the national curriculum, as well as a companion smartphone app.

Treatment 2: Unconditional cash transfer as internet data package. Households received an SMS message informing them that they would receive a free 10GB data package with 30-day validity, allowing them to opt out if they did not wish to participate. We coordinated with a large mobile provider to activate the package. This data could be used however the recipient wished. The value of this free package averaged 366 taka ($4.40 USD), which roughly equals the average per-student weekly expenditure on private tutoring (conditional on receipt) of 386 taka ($4.63 USD). We roughly estimate that the package

[7] Sample message: "Hello! Robi 10-minute school has free video lessons and quizzes to help your student keep learning! (shortened link). Text 1 if you will help your child visit the site!" Messages were delivered by SMS or voice recording (IVR).

would be sufficient for 15–20 hours of video per month.[8]

Treatment 3: Teacher support. Treated students were matched with a partner teacher from a pool of 71 teachers recruited for the study. Each recruited teacher provided a weekly, 30-minute individual phone check-in with seven assigned students for four weeks. During these meetings, teachers typically discussed students' current learning activities and plans for the week, reviewed completed work and answered student questions, and provided reviews or delivered lessons on specific topics. Teachers received a modest honorarium to cover their time and associated phone charges.

Considering that the teacher support intervention is conducted entirely remotely and provided by teachers previously unknown to students and their families, take-up of this treatment is relatively high. Slightly more than half of all invited households (54%) have a child that participates in the teacher meetings. Conditional on enrolling, students attended an average of 3.1 out of 4 meetings, with 61% of enrolled students joining all four teacher sessions.

Each treatment allows us to empirically test the net effect of reducing a different barrier to accessing remote education. Treatment 1 increases the salience of an edtech learning resource. Treatment 2 reduces the economic cost of accessing internet learning activities by providing a free data package. And Treatment 3 provides personalized teacher support to students, effectively reducing the cost of educational inputs external to the household.

In addition to these three treatments, we also measure the impact of "general information" about educational resources: we delivered information and reminders about daily TV lessons broadcast on the government satellite channel, Sangsad TV, in a similar format and frequency as the reminders about the edtech information. Because the government ceased broadcasts of regular lessons during the study period, we exclude this intervention from the main discussion.

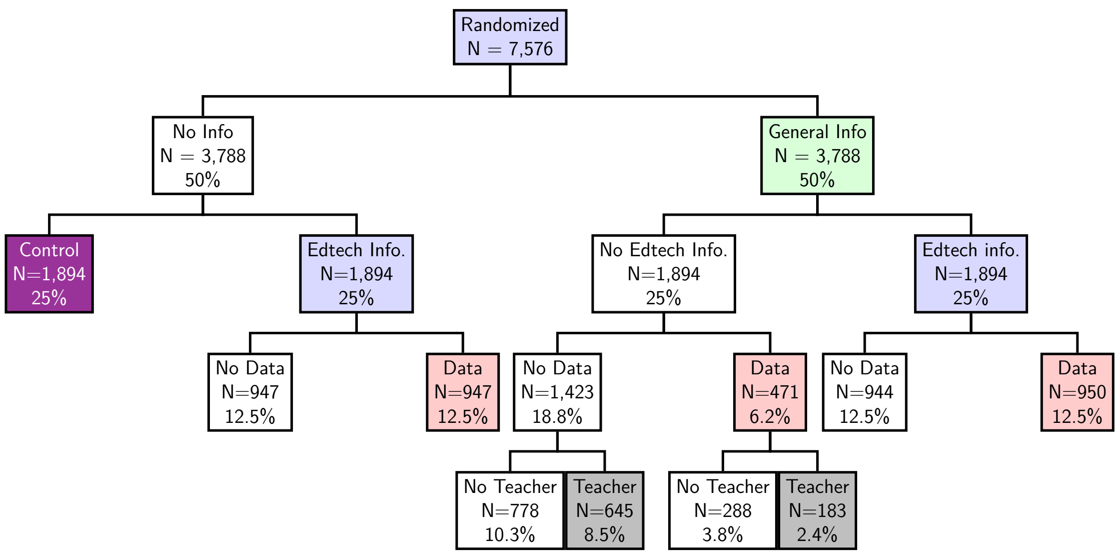
We estimate the cost per participant of delivering the information-only treatment (Treat-

[8]Calculation based on a "standard" resolution video (480p) using 480–660MB/hour (Hindy, 2022).

ment 1) as \$2.77 USD per household, which is driven mainly by fixed costs to set up the initial intervention. The total cost per participant of the messages themselves was approximately \$0.79 USD over the two months. The costs of the data package and teacher support were roughly equivalent, at \$4.40 USD and \$4.48 USD, respectively, on top of the information costs.

2.4 Randomization

Figure 1: Assignment to individual treatment arms



Notes: This figure shows the complete distribution of the treatment arms and the cross-randomizations across the three interventions: Information about an edtech tool (info), unconditional cash transfer as internet data package (data) and teacher support (teacher). It includes the share of the total and the number of participants receiving each branch.

We randomized at the household (individual-phone) level among the full set of 7,576 baseline households. Figure 1 illustrates the distribution of treatment assignments. We randomly selected half of the sample to receive Treatment 1 (edtech tool information), which was cross-randomized with the general information treatment. Treatment 2 (data package) was further cross-randomized only among those who already received some information treatment, leaving 25% of the sample to form the pure control group. Treatment 3 (teacher support) was randomized among those who received the general information treatment only. Initially, 25% of individuals in each general information only treatment arm were assigned to the teacher support treatment, but due to incomplete take-up, we expanded the share to 44% for each treatment arm. We aimed for equal splits in the cross-randomization between edtech tool information and the data package, hypothesizing that data availability could be a significant constraint for edtech tool adoption. The remaining funds were used to cross-randomize the data package with the general information treatment, resulting in a 25/75 cross-randomization split due to budget limitations.

During randomization, we stratified along four baseline dimensions: household income (five categories), sample source, child gender (whether households had male only, female only, or both male and female children in grades 6–10), and whether the household had access to at least one smartphone with an *active* internet connection.

2.5 Study timeline and data collection

We recruited and conducted a baseline survey with households by phone in September–October 2020. We targeted the caregivers of children in grades 6–10 in the household, with a nearly even split between female and male caregivers. The baseline survey included questions on demographics, family socioeconomic status, current student educational activity, parent expectations, and aspirations for their children's schooling.

We launched the three sets of interventions shortly after completing the baseline survey. We delivered informational interventions weekly for eight weeks, beginning February 24. On March 1, we distributed the initial invitation for the data package, which would last for one month. We launched the teacher support intervention simultaneously with the informational interventions, which lasted four weeks for each student.

We measure impacts on resource use and parental investments in the first follow-up survey (Round 1), conducted while the interventions were ongoing. The sampling frame comprised 7,252 baseline households that agreed to be recontacted.[9] We again targeted parents, conducting 43% of surveys with mothers, 39% with fathers, and 17% with another

[9]This reflects a 95% randomly selected subsample due to timing constraints.

family member, usually a child's older sibling. We surveyed 5,021 households, 69% of those contacted, representing 5,736 children.

We measure student learning as well as persistent impacts of the interventions in a second follow-up survey (Round 2), which took place 4 to 8 weeks after the interventions concluded. The potential sampling frame again included the set of baseline households that agreed to be recontacted, from which we attempted to contact a random subsample of 6,047 households, due to budget constraints. We successfully surveyed 3,881 households for a response rate of 55%.

During this wave, we also separately interviewed children to measure their engagement and aspirations and to assess their learning. In households with multiple children, we randomly selected one child to complete the assessment. Secondary school teachers created a bank of mathematics test questions aligned with the grade-specific national curriculum since mathematics is included in the high-stakes Secondary School Certificate (SSC) exams and is taught in all secondary grade levels and curriculum tracks.[10] The questions were designed to be asked orally and answered via multiple choice, and we piloted and revised them prior to implementation. Each student answered eight questions: a grade-specific set of four math questions at their 2020 grade level or lower, and then four additional questions at slightly lower or slightly higher grade levels, based on their performance on the initial four questions. We repeated questions across questionnaires when possible, generating a bank of 19 questions. We completed child interviews in 87% of households who completed the endline survey.

2.6 Descriptive statistics and balance tests

Column 1 of Table 1 shows the distribution of household characteristics, reported at the child level, for the entire baseline sample. Among our sample, households average 1.9

[10] We designed and implemented a similar instrument in the Bangla language subject, but because the content is not necessarily cumulative, it is difficult to differentiate student abilities across a range of grade-specific questions. Appendix C describes these challenges in more detail.

children, or 1.3 who were in grades 6–10 during the 2020 academic year. Roughly two-thirds have access to satellite or cable television, meaning that they would have the technology necessary to access lessons on the government-run television channel. Nearly all respondents were parents, with and equal distribution between mothers and fathers.

Parental education levels vary substantially, and mothers have less education on average than fathers. Specifically, 35% of mothers and 26% of fathers have completed only primary school, 18% of mothers and 17% of fathers have completed secondary school, and 18% of mothers and 25% of fathers have completed some post-secondary education.

Reflecting far lower rates of labor force participation among mothers, average mothers' unconditional income in the past 30 days is 4,864 taka ($58 USD). Income among fathers averages 51,555 taka ($619 USD), which is highly skewed relative to the median of 8,000 taka ($96 USD) per month.[11]

Parents report that their secondary school children completed school activities an average of 5.4 days per week in the month after the school closures began, which remains the same on average at the end of 2020, at 5.7 days per week.

More than half of students (59%) received private tutoring during the closures. While common globally, private supplemental tutoring is especially common in both South and East Asia (Bray, 1999; Bray and Lykins, 2012). In Bangladesh, between 68% and 81% of secondary students used private tutoring, based on various household survey estimates (Alam and Zhu, 2021), which is higher than our observed rate but comparable to the 64% of students in our sample receiving tutoring as of March 2021.

Despite concerns about the economic hardship imposed by COVID-19 pushing youth into the workforce, just 3% of youth in grades 6–10 reported working for pay in the past 30 days at baseline. These patterns of high rates of educational engagement despite the ongoing school closures are consistent with studies that focus on less advantaged populations (Beam and Mukherjee, 2024).

[11] Income is winsorized at the 99th percentile.

Table 1: Baseline descriptive statistics and balance tests

	(1)	(2)	(3)	(4)	(5)	(6)
			Baseline means			**p-value**
	All	Control	App info	Data	Teacher	Joint tests, all
Household size	1.92	1.91	1.95	1.91	1.92	0.845
	(0.99)	(0.99)	(0.95)	(0.99)	(1.02)	
Num. secondary children	1.30	1.27	1.32*	1.29	1.30	0.469
	(0.53)	(0.50)	(0.57)	(0.53)	(0.59)	
Has cable/satellite TV	0.65	0.65	0.63	0.66	0.66	0.260
	(0.48)	(0.48)	(0.48)	(0.48)	(0.47)	
Mother present	0.49	0.50	0.50	0.51	0.49	0.790
	(0.50)	(0.50)	(0.50)	(0.50)	(0.50)	
Father present	0.50	0.49	0.49	0.48	0.51	0.740
	(0.50)	(0.50)	(0.50)	(0.50)	(0.50)	
Mother primary	0.35	0.36	0.34	0.34	0.35	0.434
	(0.48)	(0.48)	(0.47)	(0.48)	(0.48)	
Mother secondary	0.18	0.17	0.20	0.19	0.18	0.395
	(0.39)	(0.38)	(0.40)	(0.39)	(0.39)	
Mother post-secondary	0.18	0.18	0.15**	0.18	0.17	0.516
	(0.38)	(0.39)	(0.36)	(0.38)	(0.38)	
Father primary	0.26	0.25	0.27	0.27	0.25	0.768
	(0.44)	(0.43)	(0.45)	(0.44)	(0.43)	
Father secondary	0.17	0.16	0.17	0.17	0.19	0.359
	(0.37)	(0.37)	(0.37)	(0.37)	(0.39)	
Father post-secondary	0.25	0.25	0.24	0.25	0.24	0.726
	(0.44)	(0.44)	(0.43)	(0.43)	(0.43)	
Mother income	4864	4550	4911	5670	3394	0.000***
	(25390)	(24830)	(24072)	(27582)	(21705)	
Father income	51555	51415	53640	50510	50834	0.726
	(134271)	(134679)	(138471)	(131624)	(130614)	
School days/week, curr.	5.70	5.76	5.66	5.69	5.64	0.917
	(2.23)	(2.17)	(2.30)	(2.23)	(2.29)	
School days/week, Apr. 20	5.37	5.38	5.40	5.37	5.43	0.923
	(2.16)	(2.18)	(2.15)	(2.16)	(2.12)	
Has private tutor	0.59	0.58	0.60	0.59	0.60	0.818
	(0.49)	(0.49)	(0.49)	(0.49)	(0.49)	
Working for pay	0.03	0.03	0.03	0.03	0.02	0.622
	(0.17)	(0.18)	(0.17)	(0.16)	(0.15)	
Number of students	8771	2175	1111	2524	954	
Number of households	7576	1894	947	2185	828	
Joint test, p-val			0.379	0.614	0.465	

Notes: Sample includes all randomized baseline respondents at the child level. Stars in columns 3–5 indicate statistically significant differences relative to the control group (column 2). Column 6 reports p-values based on F-tests of the joint significance of all treatment indicators, excluding respondents with missing values. P-values in the bottom row are from seemingly unrelated regressions that predict treatment assignment relative to the control group, with missing variable flags included. Standard errors are clustered at the household level. Stratification-cell fixed effects are included in all regressions. ***, **, and * indicate significance at the 1, 5, and 10 percent significance levels.

To test whether our sample is balanced by treatment assignment, we first test whether covariates means are equal on average between individual treatment arms and the control group. For columns 3–5, we compare all those assigned to that respective treatment arm to those in the pure control group, indicating statistically significant differences with asterisks. The final column reports p-values from testing whether all treatment variables—including interaction terms—predict each baseline characteristics.

Our sample is generally well-balanced along these pre-specified baseline covariates. Among the set of tested covariates, we only reject the null hypothesis of equal means across treatment arms in the case of mothers' income. When testing whether these covariates jointly predict treatment assignment relative to the control group using seemingly unrelated regressions, however, we do reject equal covariate means between the edtech information arm and the control group at the 10% level.

As noted earlier, we do not expect that our sample will be nationally representative of the population of households with secondary-age children. Appendix Table A.2 compares key demographic characteristics of baseline sample with households from the 2019 Multiple Indicator Cluster Survey that have a child enrolled in grades 6–10. We see that households that have below-median socioeconomic status are most comparable to the general population, with roughly equal rates of parents with post-secondary education. However, the share of parents that did not complete primary is still lower among the poorer baseline sample, at 34% and 40% for mothers and fathers, respectively, versus 43% and 50% in the overall population.

2.7 Attrition

When collecting resource usage and parental investments outcomes in the Round 1 survey, we reach 69% of households that we attempted to contact in the Round 1 survey, and treatment assignment does not predict the likelihood of recontact (Appendix Table A.3). Additionally, baseline characteristics among those who received the edtech information, information and data package, or teacher support are indistinguishable from the control group (Appendix Table A.4).

The response rate in Round 2, when we collect learning outcomes, is similar to Round 1, and we reach 65% of households. We attempted learning assessments with only one child per household, such that we completed assessments with a child in 87% of households that completed the Round 2 survey. We do find evidence that treatment assignment is associated with the likelihood of recontact and learning assessment completion (Appendix Table A.3). We therefore reject a null hypothesis of equal response rates across all treatment arms at the 5% level ($p = 0.015$) for Round 2, and at the 1% level ($p < 0.001$) for the learning assessments. In terms of respondent characteristics among Round 2 and learning assessment respondents, we do not reject equal distribution of baseline characteristics between each of our main treatment arms and the control group (Appendix Tables A.5 and A.6). We further discuss the robustness of our learning results to differential attrition in Section 5.

2.8 Empirical specification

We estimate intention-to-treat effects, reflecting the causal impact of assignment to each treatment arm on our outcomes of interest. We examine impacts across the entire sample and investigate treatment heterogeneity by socioeconomic status, which we pre-specified in our analysis plan. Because some households have more than one child in grades 6–10, we estimate our models at the child level and cluster our standard errors at the household level to reflect the household-level randomization (Abadie et al., 2017).

We estimate equations of the following general form:

$$
\begin{aligned}
y_{hc} = \alpha &+ \beta_1 EdtechInfo_h + \beta_2 Data_h + \beta_3 Teacher_h \\
&+ \beta_4 GenInfo_h + \beta_5 EdtechInfo \times GenInfo_h + \beta_6 Data_h \times EdtechInfo_h \\
&+ \beta_7 Data_h \times GenInfo_h + \beta_8 Data_h \times EdtechInfo_h \times GenInfo_h \\
&+ \beta_9 Teacher_h \times Data_h \times GenInfo_h + X_{hc}'\gamma + f_s + \epsilon_{hc}
\end{aligned}
$$

where y_{hc} is our outcome variable of interest measured at the *household-child* level. The first three binary indicators, $EdtechInfo_h$, $Data_h$, and $Teacher_h$, correspond to our main treatments of interest: Edtech information (Treatment 1), data package (Treatment 2), and teacher support (Treatment 3). We also include $GenInfo_h$, which is equal to 1 if household h receives general information about the government TV channel, and a full set of treatment-arm interactions, as specified in our pre-analysis plan (Muralidharan et al., 2019).

Following our pre-analysis plan, we include stratification cell fixed effects (f_s) and use lasso regression to select relevant covariates (Urminsky et al., 2016), selecting a penalty parameter that minimizes the 5-fold cross-validated mean squared error. Tables 2, 3, 4, and 5 show our main results, reporting the key estimated treatment coefficients of interest, $\hat{\beta}_1$, $\hat{\beta}_2$ and $\hat{\beta}_3$. Tables A.7, A.8, A.9, and A.10 report all treatment and treatment interaction coefficients. We report results that control only for stratification-cell fixed effects in Appendix Tables A.11 A.12, A.13, and A.14.

The outcome variables of interest are parent-reported measures of financial investments, time investments, and student use of technology- and non-technology-dependent learning resources (measured in Rounds 1 and 2) and student learning (measured in Round 2).[12]

In domains for which we have multiple indicators, we also generate an index based on a simple average of the component outcomes normalized to the control-group mean and standard deviation, following Kling et al. (2007).[13] For individual outcomes, we adjust for multiple hypothesis testing within each domain by reporting sharpened q-values (Anderson, 2008) alongside the p-values for our key estimated treatment coefficients of interest: $\widehat{\beta_1}$, $\widehat{\beta_2}$, and $\widehat{\beta_3}$.[14]

[12]These variables are registered in our pre-analysis plan. Appendix Tables A.15 and A.16 present the results on the other two pre-specified domains that are not central to our discussion: student engagement and time use.

[13]In the case of respondents with one or more missing outcome variables, we generate an index by averaging the remaining outcomes for which we have data.

[14]We adjust for these three coefficients only, rather than the full set of treatment interactions, because $\widehat{\beta_1}$–$\widehat{\beta_3}$ reflect the hypotheses we are testing.

2.9 Statistical power

Our primary outcomes of interest center on parental educational investment, comparing each of our main treatment arms to a control group at the Round 1 follow-up. With a significance level of 5%, we have 80% power for a minimum detectable effect size of 0.10–0.14 standard deviation impact on parental education investment when comparing each treatment arm to the control group.[15]. For an outcome like whether the likelihood of receiving private tutoring, we are powered to detect an impact of 4.8–6.4 percentage points relative to a control-group rate of 65%. For downstream impacts on learning, our power is slightly less, as we have 80% power to detect an impact of 0.18 s.d. when comparing the edtech information arm to the control group, 0.12 stanadard deviation for the data arm, and 0.16 s.d for the teacher arm. For simplicity, these calculations reflect statistical power when covariates have no predictive power, rendering them slightly conservative.

3 Descriptive insights on parents' educational inputs

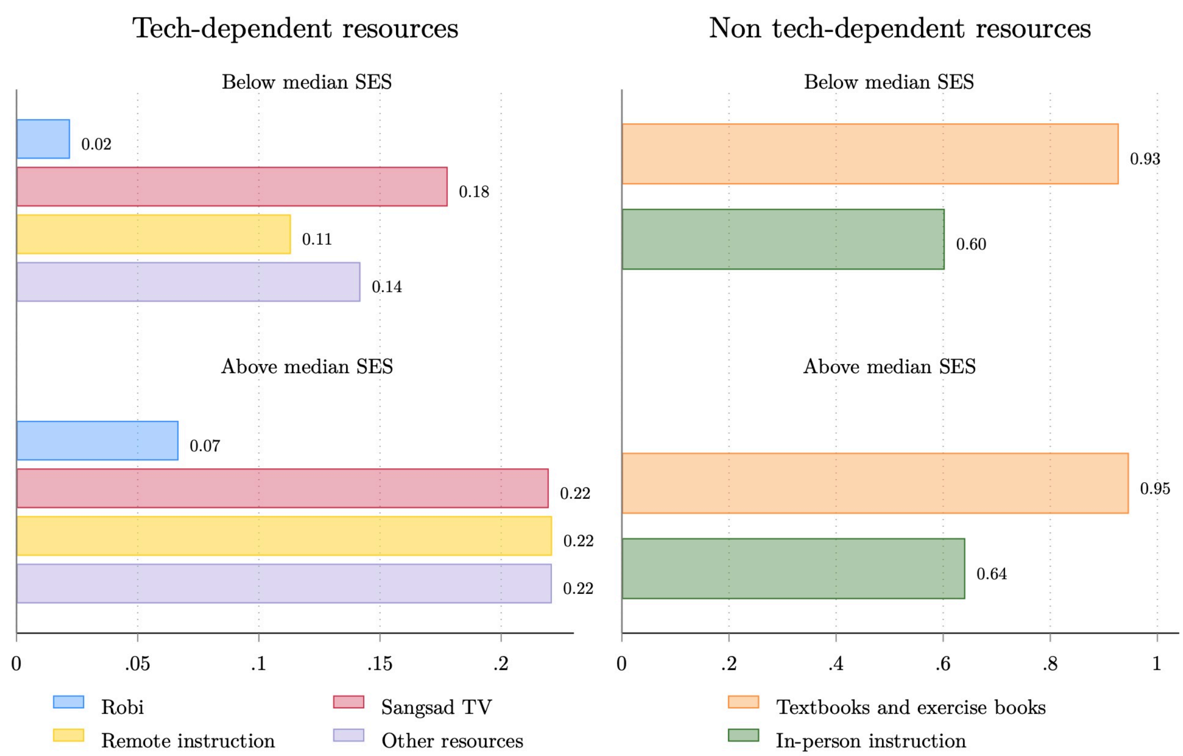
Before presenting the experimental results of the interventions aimed at relieving constraints to remote education, this section documents parental time and economic investments in children's education during the COVID-19 school closures to understand how parents invested in education when formal schooling was unavailable. We focus on three dimensions: type of learning resources used, economic investment measured by self-reported tutoring expenditures, and time investment measured by self-reported weekly hours helping their children. Additionally, we analyze socioeconomic differences in parental education investments by dividing the sample into wealthier and poorer households, based on the median of the first principal component across a series of socioeconomic status measures.[16] This

[15]The range reflects a constant control-group size of 1,408 but differing rates of assignment to treatment across each of the three arms.

[16]Following our pre-analysis plan, we take the first principal component of the following household SES measures collected at baseline: home ownership, whether members have a bank account, household asset ownership (20 items), fuel and water sources (binary indicators for each type), electricity, number of rooms

section uses investment information data from March 2021, collected while the interventions were ongoing. Hence, we examine data only from the control group to avoid confounding the descriptive evidence with treatment effects.

Figure 2: Use of learning resources



Notes: This figure reports the average use of tech-dependent and non-tech-dependent resources, disaggregated by socioeconomic status. It contains data only for the control group collected during Round 1 (March 2021). "Robi" is the name of the edtech tool promoted in Treatment 1.

Although schools had remained closed for nearly one year, rates of engagement in educational activities are high, with 89% of children doing school activities at least weekly and 78% of students studying or doing schoolwork at least 5 days on a typical week. This rate in line with a separate study of poor households in Bangladesh, among whom 79% of children in grades 8 and below reported they were doing school activities as of December 2020 (**?**). Turning globally, this rate of educational involvement appears roughly on par with LMICs

for sleeping, latrine type (binary indicators for each type), and whether there is a separate kitchen.

in Latin America and Southeast Asia. However, it is much higher than reported rates among households in Sub-Sarahan African countries (World Bank Group, 2023).

Parents average 6.6 hours per week helping their children (unconditional), or an average of 9.5 hours among the 69% who ever help. Disaggregating by type of investment, Figure 2 and Figure 3 highlight significant differences between the use of tech- and non-tech-learning resources, and that wealthier and poorer households experienced different trade-offs between time and economic investments.

Figure 2 reports the average use of tech-dependent and non-tech-dependent resources disaggregated by socioeconomic status, showing differential use of tech-dependent learning resources across both groups. Non-tech-dependent resources are widely used homogeneously, with 93–95% of students using textbooks and 60–64% meeting with an in-person teacher or tutor in the past month.

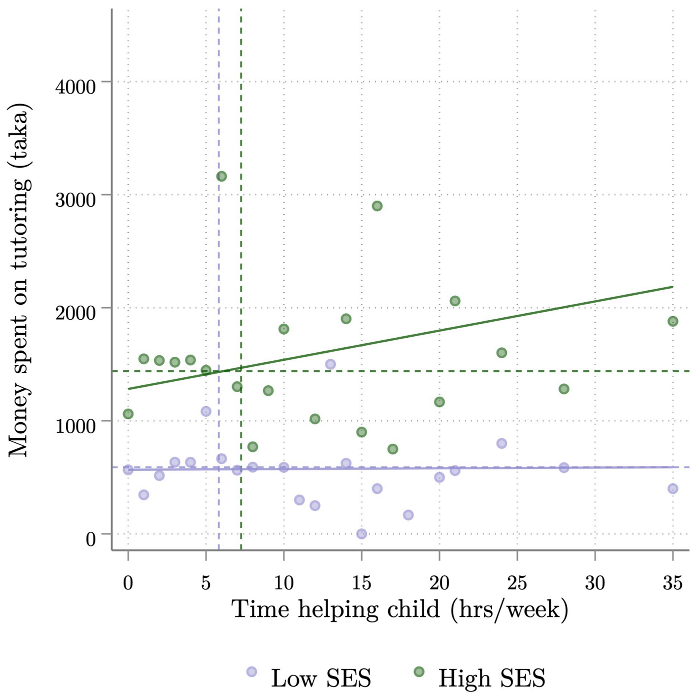
In contrast, the use of tech-dependent resources is more limited. The most popular tech-dependent resources are used by at most 22% of individuals in wealthier households. Other resources show even lower usage: government-televised lessons on Sangsad TV are used by 18–22% of students, remote teachers and classes by 11–22%, and the targeted edtech tool by only 2–7%. Additionally, tech-dependent resource use differs significantly by socioeconomic status, with higher adoption rates among students from wealthier households.[17] This suggests that economic constraints and social norms may disproportionately hinder access to edtech resources for students from poorer households.

Figure 3 shows the relationship between parental economic investment (money spent on tutoring) and time investment (time helping the child), disaggregated by socioeconomic status. It is theoretically ambiguous whether wealthier parents will invest more or less time helping their children compared to poorer parents, given the higher opportunity cost of their time and the lower cost for high-skilled parents to generate an effective unit of time

[17] Although the sample eligibility criteria required that all surveyed households had access to a smartphone, at baseline, 47% of below-median SES households and 58% of above-median households had an active data plan on their phone.

investment. Hence, the absolute number of hours invested by wealthier parents and their relative weight compared to monetary investment depend on this trade-off and on whether time and monetary investments are perceived as substitutes or complements in the human capital production function.

Figure 3: Relationship between parental time and economic investment



Notes: This figure reports the relationship between parental economic investment (money spent on tutoring) and time investment (time helping the child), disaggregated by socioeconomic status. Each data point plots the mean amount of money spent on tutoring for each (discrete) value of hours helping the child per week. It contains data only for the control group collected during Round 1 (March 2021).

In our (control) sample, parents from all socioeconomic backgrounds invest significant resources to support their children's learning. Private tutoring is very prevalent, with 64% of households reporting using it in the past month, dedicating on average 1028 taka ($12.46) per month.[18] Wealthier families are more likely to hire private tutors (68%), but 59% of children in poorer households also use this service, indicating extensive use of private tutoring across

[18]This and all other conversions are based on 1 USD = 83.28 Bangladeshi taka, the average exchange rate from April–June 2021 (OANDA, 2021).

the population, consistent with other studies (Alam and Zhu, 2021).

Parents dedicate an average of 6.5 hours a week to supporting their children with learning activities. Parents of higher socioeconomic status tend to spend slightly more time per week helping their children with schoolwork (7.25 hours/week compared to 5.82 hours/week in poorer households), consistent with Andrabi et al. (2012) in Pakistan. The primary difference between both groups is in the money spent on tutoring: Poorer households spend on average 589 taka ($7.07) per month, whereas wealthier households spend 1,439 taka ($17.28) per month. This difference could reflect variations in the number of tutoring hours used or the price per hour paid by each group.

The figure also shows that the relationship between economic and time investment is positive for wealthier households and zero for poorer households: wealthier parents who spend more money on tutoring also tend to spend more hours per week helping their children with educational activities. This finding is in line with descriptive evidence from the United States, showing that more educated parents tend to spend more time on childcare and especially on education-oriented activities (Guryan et al., 2008; Kalil et al., 2016; Ramey and Ramey, 2010; Bansak and Starr, 2021) and larger monetary investments (Corak, 2013; Kornrich and Furstenberg, 2013; Schneider et al., 2018). This suggests that parents may perceive both investments as complements in the human capital production function, or they may have an overall preference for educational investments. However, this positive relationship between time and economic investments does not appear for poorer families, suggesting binding constraints to investment.

4 Impacts on educational investments

In this section, we measure the impacts of the three remote educational interventions—providing edtech information, providing an unconditional cash transfer through a data package, and providing teacher support—on the use of tech- and non-tech-dependent learning resources and on parental economic and time investment responses. The key finding is that these interventions triggered changes in the usage of learning resources and in parental educational investments, regardless of whether they increased the take-up of the intended educational service.

Specifically, we find that providing edtech information did not increase the use of the recommended edtech tool but decreased the likelihood of using other tech-based learning resources. It also significantly impacted parental economic investment by increasing expenditures on private tutoring while marginally decreasing the time parents helped their children with educational activities. The cash transfer through a data package increased the reported use of edtech resources only when combined with information about them, and it led to an increase in the likelihood of receiving private tutoring. The teacher support caused parents to substitute away from non-tech learning resources but did not affect their use of tech-dependent learning resources, and it marginally decreased the economic investment in private tutoring.

Tables 2, 3 and 4 present the results for the full sample (Panel A), and disaggregated by the pre-specified socioeconomic dimension, reporting the effects separately for low-SES (Panel B) and high-SES households (Panel C). The bottom rows of each table report p-values from a test of whether treatment coefficients are equal between low- and high-SES households using a fully interacted model. Results that are not are explicitly described in the text as "exploratory analysis." As described in Section 2.8, the coefficients associated with each of our three main treatment arms reflect indicators for receiving these treatments. Interaction terms are included in these specifications, but for conciseness, we report them only in Appendix Tables 2–5.

4.1 Impacts on the usage of learning resources

This section examines the effects of the interventions on students' use of different learning resources, as reported by parents. These resources are broadly classified based on the delivery medium into tech-dependent (Sangsad educational TV, online video lessons, promoted edtech tool, remote individual teacher lessons, online classes, all reported in Table 2) and non-tech-dependent (textbooks, exercise books, and in-person teachers, all reported in Table 3).[19] For ease of presentation, the results of each intervention are presented jointly, referring to coefficients in both tables.

Panel A of Table 2, Column 3, shows that simply providing edtech information did not lead to sustained increase in the take-up of the edtech tool. We can reject even modest changes in usage at the 5% level [95% CI: -0.024, 0.012]. Additionally, there was no increase in the use of other tech-based resources. Notably, edtech information significantly decreased the use of the educational TV channel—Sangsad TV–by 4.9 p.p., a 25% decrease significant at the 1% level.

Exploratory analysis on the intensive margin shows an average weekly decrease of 11 minutes in educational TV use (a 30% decrease), and a 21-minute decrease in video lesson watching (a 28% decrease), both significant at the 5% level (Appendix Table A.17). Other tech-based learning resources also show negative but insignificant effects. Consistent with this reduction, exploratory analysis of self-reported data use indicates that households report a decrease in the likelihood of using a smartphone and pre-paid internet plan by 6.0 and 3.5 p.p., respectively (Appendix Table A.18).

Column 6 shows a net reduction in the overall use of tech-based resources, measured as an equally weighted index of binary indicators for whether students used each of five pre-specified tech-based learning resources, standardized to the control group. In contrast, Table 3 shows no detectable changes in the use of non-tech-based learning resources [95% CI: –0.05, 0.05] for any of the three resources or for a non-tech index averaging usage across them, normalized to the control group.[20]

[19] All these resources were pre-specified as outcomes.

[20] We deviated from the pre-analysis plan by excluding a fourth item, whether the student attends in-person classes. At the time of PAP submission, we anticipated schools reopening in early 2021, prior to at least one follow-up survey. However, they remained closed throughout the entire study period, making this indicator uninformative.

Table 2: Impact of interventions on tech-based resource use

	(1)	(2)	(3)	(4)	(5)	(6)
	Panel A. All					
	Sangsad TV	Video lessons	Edtech tool	Teacher remotely	Remote classes	Tech index
Edtech info	-0.049***	-0.029	-0.006	0.009	-0.008	-0.046*
	(0.017)	(0.019)	(0.009)	(0.015)	(0.011)	(0.024)
	[0.059*]	[1.000]	[1.000]	[1.000]	[1.000]	
Data	-0.008	0.098***	-0.015	-0.004	-0.003	0.042
	(0.030)	(0.037)	(0.013)	(0.024)	(0.019)	(0.042)
	[1.000]	[0.059*]	[1.000]	[1.000]	[1.000]	
Teacher support	-0.011	0.005	-0.010	0.021	0.007	0.010
	(0.021)	(0.023)	(0.010)	(0.019)	(0.015)	(0.032)
	[1.000]	[1.000]	[1.000]	[1.000]	[1.000]	
DV mean, control	0.20	0.25	0.05	0.12	0.08	-0.00
Observations	5715	5715	5715	5715	5715	5715
	Panel B. Low-SES Households					
	Sangsad TV	Video lessons	Edtech tool	Teacher remotely	Remote classes	Tech index
Edtech info	-0.049**	-0.027	-0.004	0.018	-0.006	-0.040
	(0.023)	(0.026)	(0.010)	(0.020)	(0.012)	(0.030)
	[0.324]	[1.000]	[1.000]	[1.000]	[1.000]	
Data	-0.016	0.129**	0.007	-0.009	0.002	0.038
	(0.043)	(0.050)	(0.019)	(0.028)	(0.020)	(0.053)
	[1.000]	[0.184]	[1.000]	[1.000]	[1.000]	
Teacher support	-0.040	-0.025	-0.006	0.024	0.022	0.002
	(0.028)	(0.030)	(0.013)	(0.025)	(0.017)	(0.044)
	[1.000]	[1.000]	[1.000]	[1.000]	[1.000]	
DV mean, control	0.18	0.20	0.02	0.08	0.04	-0.10
Observations	2881	2881	2881	2881	2881	2881
	Panel C. High-SES Households					
	Sangsad TV	Video lessons	Edtech tool	Teacher remotely	Remote classes	Tech index
Edtech info	-0.058**	-0.028	-0.004	-0.003	-0.017	-0.066*
	(0.026)	(0.029)	(0.016)	(0.023)	(0.020)	(0.038)
	[0.587]	[1.000]	[1.000]	[1.000]	[1.000]	
Data	-0.000	0.071	-0.036*	-0.002	0.003	0.044
	(0.042)	(0.054)	(0.018)	(0.039)	(0.031)	(0.065)
	[1.000]	[0.587]	[1.000]	[1.000]	[1.000]	
Teacher support	0.014	0.036	-0.016	0.016	-0.012	0.009
	(0.033)	(0.036)	(0.016)	(0.029)	(0.025)	(0.044)
	[1.000]	[1.000]	[1.000]	[1.000]	[1.000]	
DV mean, control	0.22	0.31	0.07	0.15	0.12	0.10
Observations	2834	2834	2834	2834	2834	2834
H vs. L: info	0.930	0.895	0.975	0.430	0.643	0.609
H vs. L: data	0.901	0.495	0.063*	0.814	0.862	0.990
H vs. L: teacher	0.310	0.168	0.604	0.978	0.227	0.899

Notes: Sample includes all Round 1 survey respondents. All regressions include stratification-cell fixed effects and indicators for general information, general and app information, and all treatment interactions, as well as baseline covariates selected using post-double-selection lasso with 5-fold cross-validation. Results with all treatment indicators and interactions reported in Appendix Table A.7. Robust standard errors are shown in parentheses and clustered at the household level. Anderson q-values reported in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 3: Impact of interventions on non-tech-based resource use

	(1)	(2)	(3)	(4)
Panel A. All				
	Textbooks	Exercise books	Teacher in-person	Non-Tech index
Edtech info	-0.009	0.008	0.025	-0.001
	(0.012)	(0.022)	(0.023)	(0.027)
	[1.000]	[1.000]	[0.695]	
Data	-0.007	0.039	0.018	0.038
	(0.020)	(0.037)	(0.036)	(0.050)
	[1.000]	[0.695]	[0.695]	
Teacher support	-0.030*	-0.027	-0.079***	-0.112***
	(0.016)	(0.026)	(0.027)	(0.031)
	[0.389]	[0.695]	[0.028**]	
DV mean, control	0.94	0.32	0.62	-0.00
Observations	5715	5715	5715	5715
Panel B. Low-SES Households				
	Textbooks	Exercise books	Teacher in-person	Non-Tech index
Edtech info	-0.009	-0.008	0.030	-0.001
	(0.017)	(0.033)	(0.033)	(0.038)
Data	-0.027	-0.057	0.008	-0.092
	(0.035)	(0.053)	(0.057)	(0.062)
Teacher support	-0.010	-0.024	-0.103***	-0.075*
	(0.022)	(0.036)	(0.037)	(0.043)
DV mean, control	0.93	0.32	0.60	-0.03
Observations	2881	2881	2881	2881
Panel C. High-SES Households				
	Textbooks	Exercise books	Teacher in-person	Non-Tech index
Edtech info	-0.009	0.022	0.008	-0.007
	(0.017)	(0.033)	(0.033)	(0.040)
	[0.801]	[0.760]	[1.000]	
Data	0.019	0.115**	0.036	0.170**
	(0.021)	(0.054)	(0.047)	(0.072)
	[0.760]	[0.422]	[0.760]	
Teacher support	-0.043*	-0.035	-0.059	-0.144***
	(0.024)	(0.038)	(0.039)	(0.046)
	[0.472]	[0.760]	[0.504]	
DV mean, control	0.95	0.33	0.64	0.03
Observations	2834	2834	2834	2834
H vs. L: info	0.865	0.621	0.572	0.979
H vs. L: data	0.222	0.019**	0.756	0.006***
H vs. L: teacher	0.302	0.899	0.490	0.311

Notes: Sample includes all Round 1 survey respondents . All regressions include stratification-cell fixed effects and indicators for general information, general and app information, and all treatment interactions. Results with all treatment indicators and interactions reported in Appendix Table A.8. Baseline covariates selected using post-double-selection lasso with 5-fold cross-validation. Robust standard errors are shown in parentheses and clustered at the household level. Anderson q-values reported in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The cash transfer through a data package did not significantly affect the use of edtech learning resources, allowing us to reject even small treatment effects. Additionally, there were no overall changes in tech-based or non-tech-based learning resources, as shown in Tables 2 and 3. However, disaggregating effects by learning resource shows that while the use of educational TV remained unchanged, the use of video lessons increased by 9.8 p.p. (a 39% increase), significant at the 1% level. This effect was largest for poorer households, for whom there was a 12.9 p.p. increase (65%), although we cannot reject that the coefficient is equal across socioeconomic groups ($p = 0.495$).

The data package was cross-randomized with information provision, such that 80% of recipients also received the edtech information alongside the package (Figure 1). Appendix Table A.7 shows that the data package and edtech information treatment increased the use of the recommended edtech platform by 3.5 p.p. (a 70% increase relative to 5% among the control group). This impact was starkest for wealthier households, for whom the combined edtech information and data package yielded a 4.4 p.p. increase in usage (a 62% increase). This suggests that cost was a barrier to using the edtech platform, one that was not fully surmounted for the poorest households. However, the limited usage of only 11% among the wealthiest households suggests that economic costs and knowledge about learning resources were not the only barriers families face to using edtech tools.

Consistent with this result, exploratory analysis in Appendix Table A.18 shows treatment impacts on self-reported estimated data consumption. The data package led to an 11 p.p. increase in the likelihood that respondents said they used a smartphone, a 5.6 percentage point increase in the use of a pre-paid data plan, and an estimated 2.9 GB increase in monthly data use, although the latter is not statistically significant ($p = 0.275$). However, these estimates should be interpreted with caution because measurement error is likely to be high.

Providing one-to-one phone teacher support for 30 minutes each week did not impact the extensive use of tech-based learning resources. However, it reduced the use of non-tech-based learning resources, resulting in a 0.11-SD decrease in the non-tech-dependent learning resource index (Column 3). This reduction is economically and statistically significant at the unadjusted 1% level and has an MHT-adjusted q-value of 0.004 (Table 3, Column 4). The primary driver of this decrease was a 7.9 p.p. drop in the probability of using an in-person teacher (a 13% decrease), significant at the unadjusted 1% level. Parents from both wealthier and poorer households adjusted their resource use in response to the teacher support, showing a 0.08-SD and 0.14-SD decrease in the non-tech resource index, respectively. This suggests that teacher support was seen as a direct substitute for an in-person teacher but did not affect the use of other learning resources.

Since the learning resource usage is parent-reported, the information might have increased the salience of the edtech tool, leading parents to report increased use due to social desirability or other reasons. Alternatively, students might have told parents they were using the recommended edtech tool while spending their time (and internet data) on other learning resources or distractions. However, social desirability bias seems unlikely, as we would expect an increase in reported edtech tool usage with only the provision of information. Moreover, this would not support the documented reduction in parental time spent with children as a result of edtech information. Additionally, we detect no persistent effects of the information interventions on parental investment decisions after the interventions concluded, when parents could presumably still feel pressured to report higher usage.[21]

4.2 Impacts on parental educational investments

This section reports significant impacts of the interventions on parental time investment and parental economic investment, which is measured primarily through private tutoring expenditures. The provision of information about the edtech tool and a cash transfer through the data package led parents to increase their economic investments in private tutoring, both on the extensive and intensive margins. The provision of information about the edtech tool

[21]These results are shown in Appendix Tables **??** and A.21.

additionally resulted in a drop in parental time investment, concentrated among poorer households. Conversely, the teacher support led to a modest reduction in tutoring use on the extensive margin, particularly among poorer households, suggesting that parents were substituting a more costly direct teaching with the lower-cost remote option.

Table 4 shows that the likelihood of using private tutoring increased by 4.6 p.p. after receiving the edtech tool information, a 7% increase relative to the control group mean of 64%. This result is statistically significant at the 5% unadjusted level (Panel A, Column 2), with a MHT-adjusted q-value of 0.090. On average, parents also spent 163 BDT ($1.97 USD) more per week on tutoring expenditures, a 16% increase relative to the control group mean of 1,028 BDT ($12.34 USD). This increase is statistically significant at the 5% unadjusted level and 10% adjusted level. Given that the median household pooled income in our study was 10,000 BDT per month (approximately 2,222 BDT per week), this represents a notable shift in household spending. However, about 16% of this increase in tutoring expenditures was offset by a decrease in spending on other educational resources, which declined by 27 BDT ($0.32), a 19% decline relative to the control group. In terms of parental time investment, parents' weekly hours spent helping their children study decreased by 7.5% relative to a control-group mean of 6.6 hours per week (Panel A, Column 1), although this difference was not statistically significant.

This increase in tutoring investment was observed in both wealthier and poorer households. Poorer households primarily increased spending on the intensive margin, with edtech information increasing usage by a statistically insignificant 1.6 p.p. but resulting in a 137 BDT (23%) increase in spending, significant at the unadjusted 5% level. Meanwhile, richer households increased usage by 6.2 p.p. and spending by 151 BDT (a 10% increase), with only the increase in usage being statistically significant. The reduction in parental time helping children was concentrated entirely among poorer households.

The unconditional cash transfer through a data package had similar effects to the edtech information alone, increasing tutoring investment. The likelihood of increased tutoring rose

Table 4: Impact of interventions on parental investment

	(1)	(2)	(3)	(4)
		Panel A. All		
	Hours parent helped	Private tutoring	Money on tutoring	Money on other education
Edtech info	-0.492	0.046**	163.167**	-26.870**
	(0.347)	(0.021)	(72.106)	(10.664)
	[0.186]	[0.090*]	[0.090*]	[0.090*]
Data	0.249	0.080**	149.976	18.118
	(0.716)	(0.032)	(101.620)	(18.128)
	[0.496]	[0.090*]	[0.186]	[0.368]
Teacher support	0.008	-0.047*	52.067	7.007
	(0.422)	(0.024)	(85.859)	(14.451)
	[0.697]	[0.094*]	[0.434]	[0.458]
DV mean, control	6.57	0.64	1027.82	138.56
Observations	5359	5688	5359	5065
		Panel B. Low-SES Households		
	Hours parent helped	Private tutoring	Money on tutoring	Money on other education
App. info	-1.008**	0.016	137.198**	-31.108***
	(0.442)	(0.031)	(68.635)	(10.746)
	[0.143]	[0.575]	[0.180]	[0.049**]
Data	-0.084	0.052	188.911*	40.706*
	(1.102)	(0.051)	(112.416)	(24.572)
	[0.866]	[0.360]	[0.195]	[0.195]
Teacher support	0.073	-0.064*	39.332	-9.654
	(0.543)	(0.035)	(70.636)	(14.508)
	[0.886]	[0.191]	[0.575]	[0.575]
DV mean, control	5.81	0.59	583.89	78.28
Observations	2698	2866	2735	2542
		Panel C. High-SES Households		
	Hours parent helped	Private tutoring	Money on tutoring	Money on other education
App. info	0.049	0.062**	151.463	-20.203
	(0.553)	(0.030)	(134.601)	(18.993)
	[1.000]	[0.296]	[1.000]	[1.000]
Data	0.657	0.089**	43.893	0.342
	(0.962)	(0.042)	(180.804)	(27.051)
	[1.000]	[0.296]	[1.000]	[1.000]
Teacher support	-0.022	-0.009	53.049	21.554
	(0.664)	(0.035)	(163.535)	(25.738)
	[1.000]	[1.000]	[1.000]	[1.000]
DV mean, control	7.31	0.68	1474.41	195.61
Observations	2661	2822	2624	2522
H vs. L: info	0.100	0.361	0.596	0.556
H vs. L: data	0.665	0.481	0.761	0.257
H vs. L: teacher	0.941	0.416	0.844	0.207

Notes: All expenses reported in taka. Sample includes all Round 1 survey respondents. All regressions include stratification-cell fixed effects and indicators for general information, general and app information, and treatment interactions. Results with all treatment indicators and interactions reported in Appendix Table A.9. Baseline covariates selected using post-double-selection lasso with 5-fold cross-validation. Anderson q-values reported in brackets. Robust standard errors are shown in parentheses and clustered at the household level.* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

by 8 p.p., significant at the unadjusted 5-percent level, with $q = 0.090$. Tutoring spending increased by 150 BDT ($1.80), though this estimate was imprecise and not statistically significant. Similar to the information results, both poorer and wealthier households responded economically to the intervention, with low-SES households increasing tutoring use primarily on the extensive margin, and higher-SES households increasing on the intensive margin. In contrast, there were no significant impacts on parental time investments in either sample.

Consistent with the reduction in non-tech resources shown in Table 3, teacher support via phone reduced the likelihood of students receiving private tutoring by 4.7 p.p. (a 7.3% decrease), statistically significant at the unadjusted and MHT-adjusted 10% level. There was no significant change in parents' time spent helping children. Panels B and C show that this reduction was driven by poorer households, who reduced the likelihood of private tutoring by 6.4 p.p., compared to a 0.9 p.p. reduction among wealthier households, though the difference between these groups was not statistically significant.

5 Effects on student learning

This section provides evidence that parental behavioral responses to educational interventions, rather than the interventions themselves, increased students' achievement. Edtech information improved student achievement in mathematics, with gains concentrated among wealthier households. In contrast, the cash transfer through the data package, which had a smaller effect on tutoring among high-SES households, did not affect learning. Similarly, phone teacher support did not improve math achievement. By comparing the results on parental responses to the learning results, we conclude that the increases in tutoring, especially among higher-SES parents, appears to drive the improvement in student achievement, rather than direct effects of the interventions.

Table 5 presents two measures of student achievement at endline, assessed two months after the interventions concluded. Column 1 reports the "unadjusted score," which is the

Table 5: Impact of interventions on student math learning

	(1)	(2)
Panel A. All		
	Unadjusted score	IRT, 2pl
Edtech info	0.149**	0.150**
	(0.059)	(0.058)
	[0.079*]	[0.137]
Data	-0.075	0.048
	(0.087)	(0.083)
	[1.000]	[1.000]
Teacher support	0.001	-0.016
	(0.059)	(0.058)
	[1.000]	[1.000]
DV mean, control	0.01	0.00
Observations	3433	3433
Panel B. Low-SES Households		
	Unadjusted score	IRT, 2pl
Edtech info	0.013	0.084
	(0.099)	(0.097)
	[1.000]	[1.000]
Data	-0.213	-0.153
	(0.141)	(0.132)
	[0.739]	[0.739]
Teacher support	0.014	0.057
	(0.097)	(0.090)
	[1.000]	[1.000]
DV mean, control	-0.15	-0.20
Observations	1615	1615
Panel C. High-SES Households		
	Unadjusted score	IRT, 2pl
Edtech info	0.238***	0.213***
	(0.078)	(0.075)
	[0.015**]	[0.044**]
Data	0.028	0.202*
	(0.116)	(0.110)
	[0.368]	[0.062*]
Teacher support	-0.032	-0.102
	(0.077)	(0.081)
	[0.368]	[0.096*]
DV mean, control	0.15	0.18
Observations	1808	1808
H vs. L: info	0.115	0.344
H vs. L: data	0.169	0.029**
H vs. L: teacher	0.913	0.388

Notes: Standardized score includes sum of scores on 4 math questions, normalized to the grade-specific control group. IRT adjusted score shows predicted latent ability from full set of math questions, normalized to control group mean (not grade-specific). Sample includes all Round 2 survey respondents. All regressions include stratification-cell fixed effects and indicators for general information, general and app information, and all treatment interactions. Results with all treatment indicators and interactions reported in Appendix Table A.10. Include baseline covariates selected using post-double-selection lasso with 5-fold cross-validation. Anderson q-values reported in brackets. Robust standard errors are shown in parentheses and clustered at the household level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

sum of student scores across four questions asked of all students at the same grade level, normalized to the control-group mean for each grade level. Column 2 shows impacts on predicted latent ability, based on a two-parameter item response model (IRT) applied to the full set of mathematics inventory items, normalized by the control-group mean (not grade-specific). To minimize respondent burden during the phone assessment, students were limited to eight items from the 19-question battery.[22] Consequently, these IRT-based results should be interpreted with some caution, although they closely align in magnitude and significance with the unadjusted scores based on the grade-specific base questions in Column 1 in nearly all cases.[23]

Panel A of Table 5 shows a 0.15-SD increase in mathematics scores among those who received the edtech information, whether using the unadjusted or the IRT measures in columns 1 and 2. This impact is significant at the unadjusted 5% level, with q-values of 0.079 and 0.137, respectively.

This effect is concentrated entirely among richer households, who see a 0.24-SD increase in the unadjusted measure and a 0.21-SD increase in the IRT measure, both statistically significant at the unadjusted 1% level and adjusted 5% level. There is no evidence of impacts of the edtech information among poorer students, with the 95% CI intervals of [–0.18, 0.21] using the base scores, and [–0.11, 0.27] using the 2-parameter IRT model. The concentration of math achievement gains only among high-SES students suggests that the edtech information treatment may have exacerbated existing educational inequalities.

[22] Appendix C includes a summary of descriptive statistics showing that questions have positive discrimination and capture a range of ability levels. We also observe that both the unadjusted four-question scores and latent measures are strongly correlated with student baseline ability, which we measure based on students' reported PEC math scores, the high-stakes exam students take after grade 5 (Figure A.3). In this self-reported question, students indicate whether they received an A+ (80–100), A (70–79), A- (60–69), B (50–59), C (40–49), or D (33-39).

[23] Stone (1992) finds that estimates of ability using two-parameter logistic models for test lengths of at least 10 are precise and stable using simulated data, although extreme levels of ability were biased toward zero with all tested combinations of relatively short tests (10–30 items) and relatively small samples (250–1000). Sahin and Anil (2017) use test results from university students and finds that lengths of 10 perform well conditional on a sample size of at least 750. In line with this previous work, Crawfurd et al. (2021) estimate student ability measured through a phone survey using a two-parameter model with 11–12 questions per respondent.

While the data package also increased private tutoring use, we do not see evidence of learning gains, with an impact of -0.075 SD on the unadjusted index and 0.048 SD on the IRT measure. However, disaggregating by socioeconomic status in panels B and C reveals stark differences in impacts. The coefficients for poorer households are negative and imprecise, whereas those for wealthier households are positive and, in the case of the IRT measure in column 2, quite large (0.21 SD) and comparable to the results of the information intervention. Although we cannot definitively isolate the reasons for these differences, the nature of the tutoring investment impacts may provide an explanation: the information intervention led to a large increase on the intensive margin, while the data package led to a more extensive-margin increase.

The bottom row of Table 5 indicates that the phone teacher support treatment did not significantly increase student mathematics achievement relative to the control group, neither on the grade-specific set of four "base" questions nor on estimated latent ability. This lack of impact is likely due to the light-touch nature of the teaching support, as successful teaching support interventions typically rely on more intensive treatments. Furthermore, as shown in Tables 3 and 4, parents reduced their own educational investments in their children while teacher support was provided, potentially offsetting any gains.

These results should be interpreted with some caution for two reasons. First, assessments conducted by phone limited the depth and breadth of questions asked. Second, unlike the initial follow-up round (Round 1), there is evidence of differential attrition by treatment assignment among those who completed the learning assessment (Round 2). However, three sets of evidence suggest our results are not driven by differences in the composition of the endline sample.

First, we demonstrate that the parental investment results are similar in magnitude when restricting to the learning assessment sample, as seen in Appendix Tables A.22, A.23, and A.24, although there is some loss of significance due to the reduced sample size. Thus, the intervention-induced changes in parental tutoring are present in this learning assessment sample. Second, Appendix Table A.25, presents our learning estimates, adjusted using inverse-propensity weighting to account for attrition using covariates in Table 1 (Busso et al., 2014; Hirano et al., 2003). The impact of edtech information on learning remains nearly identical in magnitude and statistical significance.

Finally, we follow Behaghel et al. (2015) and trim our sample based on the number of call attempts, equalizing the response rate conditional on the number of attempts between groups, focusing on the impact of edtech information. This procedure assumes that while treatment may influence a respondent's decision to participate, it does not affect their reluctance to respond *relative* to others in the same treatment arm.[24] In the case of the information treatment relative to the control group treatment, the response rates are 44.7% in the edtech information group and 44.4% in the control group after 3 and 5 attempts, respectively. We trim the sample at this point, with the results shown in Appendix Table A.26. Estimating the impacts of the edtech information with our trimmed sample yields nearly identical estimates in terms of magnitude and statistical significance.

6 Discussion

This section discusses the potential mechanisms by which the three interventions may influence parental economic and time investment and, in turn, affect children's learning and human capital development. Appendix B includes a stylized model that formalizes some of the discussions below.[25]

Edtech learning resources serve as a complement or alternative to traditional pedagogical methods and resources, implemented both in formal schooling environments and for remote education. The latter is especially important for learners from disadvantaged backgrounds,

[24]Because the discrete number of phone attempts made it unlikely that we can make response rates equal across all treatment arms, we focus on the edtech information arm.

[25]In the model, human capital is provided through internal teaching—by parents themselves—or through external teaching via schools and tutors. It characterizes the optimal household decision rules in this environment and introduces a novel form of acquiring human capital through costly education technology, where perceived and actual returns may not be equal.

who may lack access to adequate learning environments and quality instruction (Lai et al., 2012; Caballero Montoya et al., 2021) or experience higher rates of education disruption. However, edtech learning resources can have non-negligible economic costs, for the product itself, necessary devices, or technical capabilities such as internet connections or cable TV subscriptions. Other barriers to access, such as the need for technological literacy, can also hinder adoption among underprivileged groups.

Common interventions to address these barriers to edtech take-up include providing information and reducing costs.[26] Providing information can signal the value of a new platform, leading individuals to revise their beliefs about its marginal returns and potentially broadly shifting beliefs about the value of features highlighted by that technology. For example, information about an edtech tool that promotes adaptive learning methods may lead households to value adaptive learning methods in general. Additionally, edtech can make educational investment more salient, prompting parents to increase their educational investments. Hence, information about an edtech tool may have both direct impacts on tool usage and indirect impacts on general education investment. Reducing edtech costs via cash subsidies may increase usage, with the net impact depending on the perceived returns to the technology and whether the cash subsidy is directly tied to usage.

Our findings are consistent with the hypothesis that providing information about an edtech tool indirectly impacts educational investment, and that salience of investment may be an important channel. Although information about the edtech tool did not increase its use, it led to significant shifts in parental investments through a substantial increase in tutoring expenditures. This increase in tutoring expenditures occurred while information provision was ongoing, but it faded out by Round 2 survey, which took place approximately

[26]Behavioral nudges have been shown to encourage continuous investment among students enrolled in MOOC courses, where drop-out rates are high (see Yeomans and Reich (2017); Martinez (2014); Patterson (2018); Baker et al. (2016) among others). In Uruguay, e-messages and nudges targeting different behavioral biases boosted parental investment in early childhood development (Bloomfield et al., 2022). In the United States, SMS reminders for parents to do home literacy activities increased early literacy of preschoolers (York et al., 2018) and kindergarteners only when messages were personalized (Doss et al., 2019). Similar reminders for parents of children of Head Start centers (Hurwitz et al., 2015) and parents of primary school children over the summer break (Kraft and Monti-Nussbaum, 2017) also had positive effects.

two months after the interventions concluded (results shown in Appendix Table A.21). Rates of private tutoring and average tutoring expenses fell between the first and second follow-up, which suggests that the increases in tutoring relied on consistent reminders, rather than the other explanations such as the control group rates of investment caught up over time.

Although we can not conclusively identify *why* edtech reminders drove increases in tutoring, there are several possibilities. The edtech tool's focus on adaptive learning features may have highlighted the importance of personalized learning more broadly. Consequently, parents may have chosen to invest in the most familiar and available personalized learning resource: tutoring.

Households facing additional constraints might not adopt the educational technology, even if they perceive a high return on investment. Instead, they might respond to the informational message to re-optimize their investments and subsequent learning option choices *without* actually adopting the new technology. Another possibility is that tutoring may be preferred over the edtech tool in general because parents feel they have more control over their children's time (Gallego et al., 2020) or are uncertain about using novel resources effectively. Lastly, parents might simply believe that tutoring offers a higher return on investment relative to the recommended app, and the information served as a general reminder to invest. Indeed, the edtech-info-induced increase in tutoring led to learning gains, concentrated among students from wealthier backgrounds.

Despite significant shifts in other investments, there was no increase in the use of the targeted edtech tool due to the information alone, indicating the presence of other barriers, such as economic constraints. The increase in edtech tool take-up *only* occurred when the information was accompanied by reduced data costs, suggesting that parents were more willing to invest in the novel remote learning resource when costs were mitigated. This data-induced change in edtech tool usage also led to an extensive-margin increase in the use of tech-dependent options rather than substituting them. However, even with the cash subsidy, reported edtech tool take-up remained low, suggesting that other barriers persisted or that parents perceived relatively low marginal returns from the edtech tool. More specific information or content may be required to update parents' beliefs about its returns, or parents may accurately attribute low value to the resource, either because it provides intrinsically low value or because of lack of computer literacy or accompanying distractions of internet use (Beuermann et al., 2015; Piper et al., 2016; Cristia et al., 2017; Malamud et al., 2019).

The results indicate that parents adjust their economic investment relatively more than their time investment. This suggests either that parents perceive higher marginal returns from economic investments, or that time spent with their children is more inelastic in the short run than economic investment in tutoring. The heterogeneity results support the latter argument, showing that the increased tutoring effects are concentrated among wealthier families.

The increases in tutoring and learning, particularly among wealthier households, due to the edtech information suggests that tutoring itself drives the learning gains. This is further supported by the absence of changes in other educational inputs. Another possibility is that the information affected student engagement and motivation, either directly or indirectly, as a result of the observed changes in parental investments. However, we find that student effort and aspirations remain high and unaffected by the interventions (Appendix Table A.16).

The absence of learning impacts from the remote teacher support intervention aligns with findings by Crawfurd et al. (2023), who reported that a similar intervention in Sierra Leone increased engagement but not learning.[27] These results could indicate that this method of supporting student learning may not be effective. In our study, teacher support covered a range of topics, not necessarily mathematics, which may have dispersed any learning gains. Another hypothesis is that parents temporarily reallocated their investments due to the support, leading to no net change in educational investments and, consequently, no change in student engagement. Despite the short duration of the teacher support intervention, we observe a decrease in private tutoring in Table 4, indicating this reallocation.

[27] Unlike Crawfurd et al. (2023), we do not observe changes in student engagement or effort (Appendix Table A.16).

7 Conclusions

We conducted a field experiment with households across Bangladesh during the COVID-19 school closures to document parental investments and assess the impact of three short-term educational interventions designed to address different barriers to parental educational investment. The results indicate that these interventions did not directly improve learning outcomes. Information about a learning app did not lead households to use the app, and while providing information alongside a data subsidy did increase usage, the magnitude of this increase was too modest to yield detectable learning impacts. While a high share of targeted children did participate in phone-based tutoring, we do not observe evidence of learning impacts. One possibility is that the duration—4 weeks— was shorter compared to more successful phone-based tutoring interventions (Angrist et al., 2022, 2023), which provided weekly phone-based tutoring sessions for a period of 8–16 weeks. However, the most substantial difference is that in this context, teachers were allowed substantial flexibility in the curriculum and choice of subject to maximize the chance that students would remain engaged, while Angrist et al. (2022) and Angrist et al. (2022) developed a differentiated but targeted mathematics curriculum centered on a few key subjects.

However, providing educational services in the presence of other barriers may lead parents to reallocate their educational investments, potentially resulting in lasting effects on student achievement. The varying impacts of these interventions on poorer and wealthier households suggest that some policies intended to reduce barriers to accessing remote education may actually exacerbate educational inequalities.

We find that two light-touch interventions—an informational campaign promoting an edtech tool and a data package providing free internet access— triggered significant parental behavioral responses, including increases in economic investment in tutoring. The substantial increase in tutoring driven by the edtech information appears to improve learning, particularly among wealthier households. The information campaign alone does not increase usage of the recommended edtech tool; usage only rises when information is combined with the data package. Despite addressing economic barriers through the data package, usage remains low even after receiving both interventions, indicating that other constraints continue to limit edtech tool adoption.

We also find that remote individualized teacher support affects parental educational investments, leading to a reduction in private tutoring, especially among low-SES households. This reallocation could partly explain the lack of evidence for improved student mathematics learning. Other factors may include the relatively short treatment duration and the flexibility teachers had to support learning in the subjects students requested.

Due to the relatively short duration of the intervention, it is plausible that more time is needed for stronger behavioral responses. Adjusting time and economic investments can be costly, potentially more so for poorer households. Households may have anticipated a return to pre-intervention levels of internet data and one-to-one teaching support, leading them to favor their baseline investment choices. This suggests that longer interventions might produce larger and lasting changes in household investments. Additionally, the lack of detectable learning impacts from the data package may be due to low take-up and usage of the edtech tool, indicating that the edtech tool itself could be effective if used more widely.

Our results highlight the importance of parental behavioral responses in driving the impacts of educational policies and demonstrate how these decisions affect the distributional effects of such policies. The specific trade-offs and constraints parents face may be context-specific, especially given that this study was conducted during the COVID-19 pandemic, when drastically altered educational investments due to school closures. However, the core insight—that parental investment decisions, far from second-order, generate measurable and heterogeneous impacts on student outcomes—is relevant beyond this context and applies to various settings (Das et al., 2013). Furthermore, our findings indicate that parents value personalized support, such as private tutoring, which has detectable effects on student learning. While phone data can limit the take-up and use of edtech tools, its unconstrained provision alone does not produce learning gains, highlighting the need for additional guidance and personalized support to fully realize the benefits of educational technologies.

From a policy perspective, these results highlight the potential of remotely delivered interventions to affect parental educational investments and promote student learning. Our findings, which show learning gains through tutoring, align with existing literature on teaching at the right level (Banerjee et al., 2007, 2016; Muralidharan et al., 2019). The effectiveness of educational policies and their implications for inequality will depend on how well policy-makers account for the role of parents' adjustment responses and potential constraints to intervention uptake when designing these policies.

References

Abadie, A., S. Athey, G. Imbens, and J. Wooldridge (2017). When Should You Adjust Standard Errors for Clustering? Working Paper w24003, National Bureau of Economic Research, Cambridge, MA.

Agostinelli, F., M. Doepke, G. Sorrenti, and F. Zilibotti (2022, February). When the great equalizer shuts down: Schools, peers, and parents in pandemic times. *Journal of Public Economics 206*, 104574.

Ahmed, S. I., M. R. Haque, J. Chen, and N. Dell (2017). Digital privacy challenges with shared mobile phone use in Bangladesh. *Proceedings of the ACM on Human-Computer Interaction 1*(CSCW), 1–20.

Alam, M. B. and Z. Zhu (2021, January). Private tutoring in Bangladesh: Evolution, expansion, and policy responses. *International Journal of Comparative Education and Development 24*(1), 20–36.

Alamgir, M. (2020). Assignments for class 6-9 students to start on Nov 1. https://www.thedailystar.net/frontpage/news/assignments-class-6-9-students-start-nov-1-1983217.

Anderson, M. L. (2008). Multiple inference and gender differences in the effects of early intervention: A reevaluation of the abecedarian, perry preschool, and early training projects. *Journal of the American Statistical Association 108*(484), 1481–1495.

Andrabi, T., B. Daniels, and J. Das (2021). Human Capital Accumulation and Disasters: Evidence from the Pakistan Earthquake of 2005. *The Journal of Human Resources*, 0520–10887R1.

Andrabi, T., J. Das, and A. I. Khwaja (2012, October). What Did You Do All Day?: Maternal Education and Child Outcomes. *Journal of Human Resources 47*(4), 873–912.

Andrew, A., S. Cattan, M. Costa Dias, C. Farquharson, L. Kraftman, S. Krutikova, A. Phimister, and A. Sevilla (2020). Inequalities in children's experiences of home learning during the COVID-19 lockdown in England. *Fiscal Studies 41*(3), 653–683.

Angrist, N., M. Ainomugisha, S. P. Bathena, P. Bergman, C. Crossley, C. Cullen, T. Letsomo, M. Matsheng, R. Marlon Panti, S. Sabarwal, and T. Sullivan (2023, May). Building Resilient Education Systems: Evidence from Large-Scale Randomized Trials in Five Countries.

Angrist, N., P. Bergman, and M. Matsheng (2022, June). Experimental evidence on learning using low-tech when school is out. *Nature Human Behaviour*, 1–10.

Attanasio, O., T. Boneva, and C. Rauh (2020, September). Parental Beliefs about Returns to Different Types of Investments in School Children. *The Journal of Human Resources*, 0719.

Axiata Group Berhad (2020). Integrated Annual Report 2020. Annual Report.

Bacher-Hicks, A., J. Goodman, and C. Mulhern (2021). Inequality in household adaptation to schooling shocks: Covid-induced online learning engagement in real time. *Journal of Public Economics 193*.

Baker, R., B. Evans, and T. Dee (2016). A Randomized Experiment Testing the Efficacy of a Scheduling Nudge in a Massive Open Online Course (MOOC). *AERA Open 2*(4).

Bandiera, O., N. Buehren, M. Goldstein, I. Rasul, and A. Smurra (2020). Do School Closures During an Epidemic have Persistent Effects? Evidence from Sierra Leone in the Time of Ebola. Working Paper.

Banerjee, A., R. Banerji, J. Berry, E. Duflo, H. Kannan, S. Mukherji, M. Shotland, and M. Walton (2016). Mainstreaming an effective intervention: Evidence from randomized evaluations of "Teaching at the Right Level" in India. Technical report, National Bureau of Economic Research.

Banerjee, A. V., S. Cole, E. Duflo, and L. Linden (2007). Remedying education: Evidence from two randomized experiments in India. *The Quarterly Journal of Economics 122*(3), 1235–1264.

Bansak, C. and M. Starr (2021). COVID-19 shocks to education supply: How 200,000 US households dealt with the sudden shift to distance learning. *Review of Economics of the Household 19*(1), 63–90.

Beam, E. and P. Mukherjee (2024). Gender-specific Earnings Losses and Child Well-being: Evidence from vulnerable households in Bangladesh. Working Paper.

Becker, G. S. (1965). A Theory of the Allocation of Time. *The Economic Journal 75*(299), 493–517.

Becker, G. S. and N. Tomes (1976). Child Endowments and the Quantity and Quality of Children. *Journal of Political Economy 84*(4), S143–S162.

Beg, S. A., A. M. Lucas, W. Halim, and U. Saif (2019). Beyond the Basics: Improving Post-Primary Content Delivery Through Classroom Technology. *National Bureau of Economic Research* (Working Paper 25704).

Behaghel, L., B. Crépon, M. Gurgand, and T. Le Barbanchon (2015). Please call again: Correcting nonresponse bias in treatment effect models. *Review of Economics and Statistics 97*(5), 1070–1080.

Beuermann, D. W., J. Cristia, S. Cueto, O. Malamud, and Y. Cruz-Aguayo (2015). One laptop per child at home: Short-term impacts from a randomized experiment in Peru. *American Economic Journal: Applied Economics 7*(2), 53–80.

Blanden, J., M. Doepke, and J. Stuhler (2022). Educational inequality. Technical report, National Bureau of Economic Research.

Bloomfield, J., A. Balsa, and A. Cid (2022). Using behavioral insights in early childhood interventions: The effects of Crianza Positiva e-messaging program on parental investment. *Review of Economics of the Household*, 1–36.

Boneva, T. and C. Rauh (2018). Parental beliefs about returns to educational investments—the later the better? *Journal of the European Economic Association 16*(6), 1669–1711.

Bray, M. and C. Lykins (2012, May). *Shadow Education: Private Supplementary Tutoring and Its Implications for Policy Makers in Asia.* Asian Development Bank.

Bray, T. M. (1999). *The Shadow Education System: Private Tutoring and Its Implications for Planners.* UNESCO International Institute for Educational Planning.

Busso, M., J. DiNardo, and J. McCrary (2014). New evidence on the finite sample properties of propensity score reweighting and matching estimators. *Review of Economics and Statistics 96*(5), 885–897.

Caballero Montoya, E., D. Mtambo, K. Nzomo, M. Shotland, and J. Thunde (2021). Distance Learning. What have we learned? Technical report, IDinsight.

Carlana, M. and E. La Ferrara (2021). Apart but Connected: Online Tutoring and Student Outcomes during the COVID-19 Pandemic. *SSRN Electronic Journal*.

Caucutt, E. M., L. Lochner, and Y. Park (2017). Correlation, consumption, confusion, or constraints: Why do poor children perform so poorly? *The Scandinavian Journal of Economics 119*(1), 102–147.

Corak, M. (2013, September). Income Inequality, Equality of Opportunity, and Intergenerational Mobility. *Journal of Economic Perspectives 27*(3), 79–102.

Crawfurd, L., D. K. Evans, S. Hares, and J. Sandefur (2021). *Teaching and Testing by Phone in a Pandemic*. Center for Global Development.

Crawfurd, L., D. K. Evans, S. Hares, and J. Sandefur (2023, September). Live tutoring calls did not improve learning during the COVID-19 pandemic in Sierra Leone. *Journal of Development Economics 164*, 103114.

Cristia, J., P. Ibarrarán, S. Cueto, A. Santiago, and E. Severín (2017). Technology and child development: Evidence from the one laptop per child program. *American Economic Journal: Applied Economics 9*(3), 295–320.

Cunha, F., J. J. Heckman, L. Lochner, and D. V. Masterov (2006). Chapter 12 Interpreting the Evidence on Life Cycle Skill Formation. In E. Hanushek and F. Welch (Eds.), *Handbook of the Economics of Education*, Volume 1, pp. 697–812. Elsevier.

Dahl, G. B. and L. Lochner (2012). The impact of family income on child achievement: Evidence from the earned income tax credit. *American Economic Review 102*(5), 1927–56.

Das, J., S. Dercon, J. Habyarimana, P. Krishnan, K. Muralidharan, and V. Sundararaman (2013). School Inputs, Household Substitution, and Test Scores. *American Economic Journal: Applied Economics 5*(2), 29–57.

Del Bono, E., L. Fumagalli, A. Holford, and B. Rabe (2021). Coping with school closures: Changes in home-schooling during COVID-19. *Institute for Economic and Social Research Report*.

Dizon-Ross, R. (2019). Parents' Beliefs about Their Children's Academic Ability: Implications for Educational Investments. *American Economic Review 109*(8), 2728–2765.

Doss, C., E. M. Fahle, S. Loeb, and B. N. York (2019). More than just a nudge supporting kindergarten parents with differentiated and personalized text messages. *The Journal of Human Resources 54*(3), 567–603.

Francesconi, M. and J. J. Heckman (2016). Child Development and Parental Investment: Introduction. *The Economic Journal 126*(596), F1–F27.

Fredriksson, P., B. Ockert, and H. Oosterbeek (2016, October). Parental Responses to Public Investments in Children: Evidence from a Maximum Class Size Rule. *The Journal of Human Resources 51*(4), 832–868.

Fuchs-Schündeln, N., D. Krueger, A. Ludwig, and I. Popova (2022). The long-term distributional and welfare effects of covid-19 school closures. *The Economic Journal 132*(645), 1647–1683.

Gallego, F. A., O. Malamud, and C. Pop-Eleches (2020). Parental monitoring and children's internet use: The role of information, control, and cues. *Journal of Public Economics 188*, 104208.

Gelber, A. and A. Isen (2013). Children's schooling and parents' behavior: Evidence from the Head Start Impact Study. *Journal of Public Economics 101*, 25–38.

Glewwe, P., Z. Li, P. Loyalka, K. W. Rahman, and U. Sharma (2024). Education without formal schooling through tablets and tutors: Evidence from out-of-school children in bangladesh during the covid-19 pandemic. *Asian Development Bank Economics Working Paper Series* (718).

Guryan, J., E. Hurst, and M. Kearney (2008, September). Parental Education and Parental Time with Children. *Journal of Economic Perspectives 22*(3), 23–46.

Hassan, H., A. Islam, A. Siddique, and L. C. Wang (2021). Telementoring and homeschooling during school closures: A randomized experiment in rural Bangladesh.

Hindy, J. (2022, May). How much data does YouTube actually use? https://www.androidauthority.com/how-much-data-does-youtube-use-964560/.

Hirano, K., G. W. Imbens, and G. Ridder (2003). Efficient estimation of average treatment effects using the estimated propensity score. *Econometrica 71*(4), 1161–1189.

Houtenville, A. J. and K. S. Conway (2008). Parental Effort, School Resources, and Student Achievement. *The Journal of Human Resources 43*(2), 437–453.

Hurwitz, L. B., A. R. Lauricella, A. Hanson, A. Raden, and E. Wartella (2015). Supporting Head Start parents: Impact of a text message intervention on parent–child activity engagement. *Early Child Development and Care 185*(9), 1373–1389.

Johnston, J. and C. Ksoll (2017). Effectiveness of Interactive Satellite-Transmitted Instruction: Experimental Evidence from Ghanaian Primary Schools. *Center for Education Policy Analysis Working Paper No. 17-08*.

Kalil, A., K. M. Ziol-Guest, R. M. Ryan, and A. J. Markowitz (2016, July). Changes in Income-Based Gaps in Parent Activities With Young Children From 1988 to 2012. *AERA Open 2*(3), 2332858416653732.

Kling, J. R., J. B. Liebman, and L. F. Katz (2007). Experimental Analysis of Neighborhood Effects. *Econometrica 75*(1), 83–119.

Kornrich, S. and F. Furstenberg (2013). Investing in Children: Changes in Parental Spending on Children, 1972—2007. *Demography 50*(1), 1–23.

Kraft, M. A. and M. Monti-Nussbaum (2017). Can Schools Enable Parents to Prevent Summer Learning Loss? A Text-Messaging Field Experiment to Promote Literacy Skills. *The ANNALS of the American Academy of Political and Social Science 674*(1), 85–112.

Lai, F., L. Zhang, Q. Qu, and X. Hu (2012). Does Computer-Assisted Learning Improve Learning Outcomes? Evidence from a Randomized Experiment in Public Schools in Rural Minority Areas in Qinghai, China.

Lichand, G., C. A. Doria, O. Leal-Neto, and J. P. C. Fernandes (2022). The impacts of remote learning in secondary education during the pandemic in brazil. *Nature Human Behaviour*, 1–8.

List, J. A., J. Pernaudet, and D. Suskind (2021, October). It All Starts with Beliefs: Addressing the Roots of Educational Inequities by Shifting Parental Beliefs.

Malamud, O., S. Cueto, J. Cristia, and D. W. Beuermann (2019, May). Do children benefit from internet access? Experimental evidence from Peru. *Journal of Development Economics 138*, 41–56.

Martinez, I. (2014). Never Put Off Till Tomorrow?

Muralidharan, K., A. Singh, and A. J. Ganimian (2019). Disrupting education? Experimental evidence on technology-aided instruction in India. *American Economic Review 109*(4), 1426–60.

Navarro-Sola, L. (2021). Secondary schools with televised lessons: The labor market returns of the Mexican telesecundaria.

OANDA (2021). Historical exchange rates. Accessed September 20, 2021.

Okeleke, K. (2021). Achieving mobile-enabled digital inclusion in Bangladesh.

Patterson, R. W. (2018). Can behavioral tools improve online student outcomes? Experimental evidence from a massive open online course. *Journal of Economic Behavior & Organization 153*, 293–321.

Piper, B., S. S. Zuilkowski, D. Kwayumba, and C. Strigel (2016, July). Does technology improve reading outcomes? Comparing the effectiveness and cost-effectiveness of ICT interventions for early grade reading in Kenya. *International Journal of Educational Development 49*, 204–214.

Pop-Eleches, C. and M. Urquiola (2013). Going to a Better School: Effects and Behavioral Responses. *American Economic Review 103*(4), 1289–1324.

Ramey, G. and V. A. Ramey (2010, March). The Rug Rat Race. Technical report.

Sahin, A. and D. Anil (2017). The Effects of Test Length and Sample Size on Item Parameters in Item Response Theory.

Schneider, D., O. P. Hastings, and J. LaBriola (2018, June). Income Inequality and Class Divides in Parental Investments. *American Sociological Review 83*(3), 475–507.

Schueler, B. E. and D. Rodriguez-Segura (2021). A Cautionary Tale of Tutoring Hard-to-Reach Students in Kenya.

Singh, A., M. Romero, and K. Muralidharan (2022). COVID-19 Learning Loss and Recovery: Panel Data Evidence from India. Technical Report 22/122.

Stone, C. A. (1992, March). Recovery of Marginal Maximum Likelihood Estimates in the Two-Parameter Logistic Response Model: An Evaluation of MULTILOG. *Applied Psychological Measurement 16*(1), 1–16.

Todd, P. E. and K. I. Wolpin (2003). On the Specification and Estimation of the Production Function for Cognitive Achievement. *The Economic Journal 113*(485), F3–F33.

Todd, P. E. and K. I. Wolpin (2007). The Production of Cognitive Achievement in Children: Home, School, and Racial Test Score Gaps. *Journal of Human Capital 1*(1), 91–136.

UNESCO (2022). Global monitoring of school closures caused by COVID-19. https://en.unesco.org/covid19/educationresponse.

Urminsky, O., C. Hansen, and V. Chernozhukov (2016). Using double-lasso regression for principled variable selection. *Available at SSRN 2733374*.

Wang, L. C., M. Vlassopoulos, A. Islam, and H. Hassan (2023). Delivering remote learning using a low-tech solution: Evidence from a randomized controlled trial in bangladesh.

World Bank Group (2023). Covid-19 high frequency monitoring dashboard. https://www.worldbank.org/en/data/interactive/2020/11/11/covid-19-high-frequency-monitoring-dashboard. World Bank Group, Washington, DC. Accessed: 2024-10-17.

Yeomans, M. and J. Reich (2017). Planning prompts increase and forecast course completion in massive open online courses. In *Proceedings of the Seventh International Learning Analytics & Knowledge Conference*, Vancouver British Columbia Canada, pp. 464–473. ACM.

York, B. N., S. Loeb, and C. Doss (2018). One Step at a Time: The Effects of an Early Literacy Text Messaging Program for Parents of Preschoolers. *The Journal of Human Resources*, 0517.

A For Online Publication: Appendix Figures and Tables

Figure A.1: Project timeline

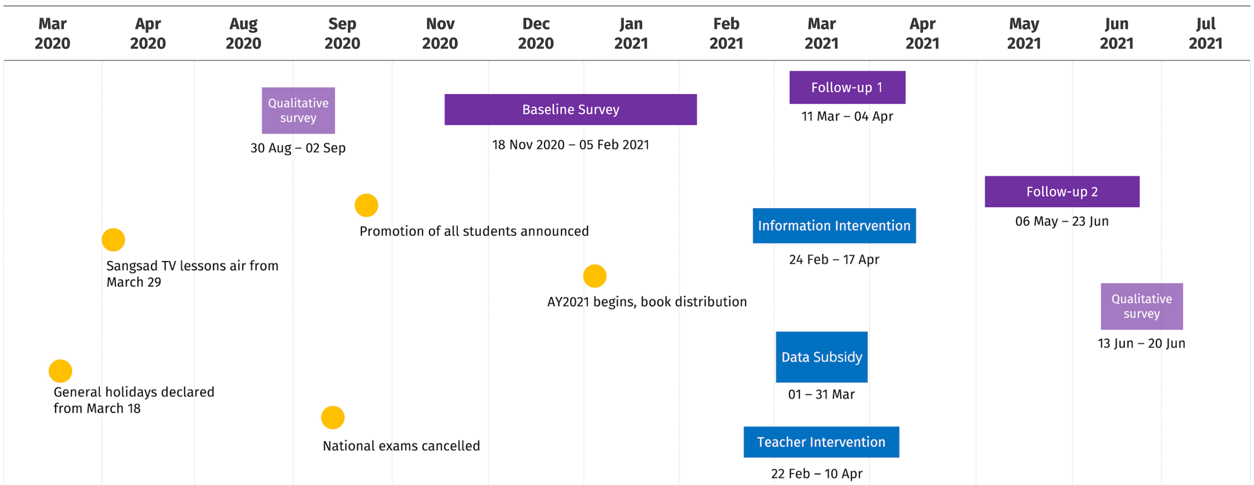


Figure A.2: Distribution of respondents

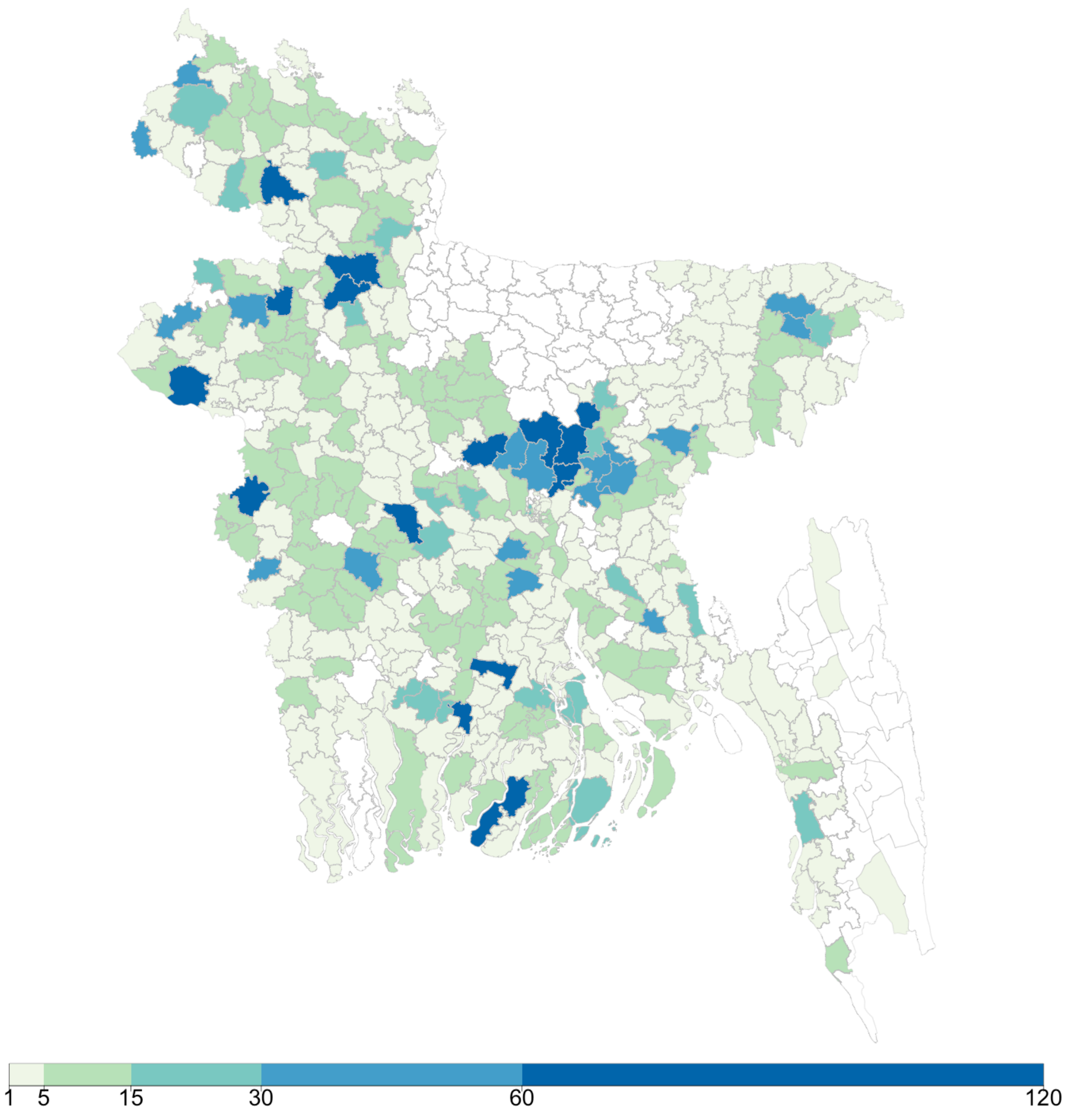


Figure A.3: Distribution of endline math scores by self-reported Grade 5 exam scores

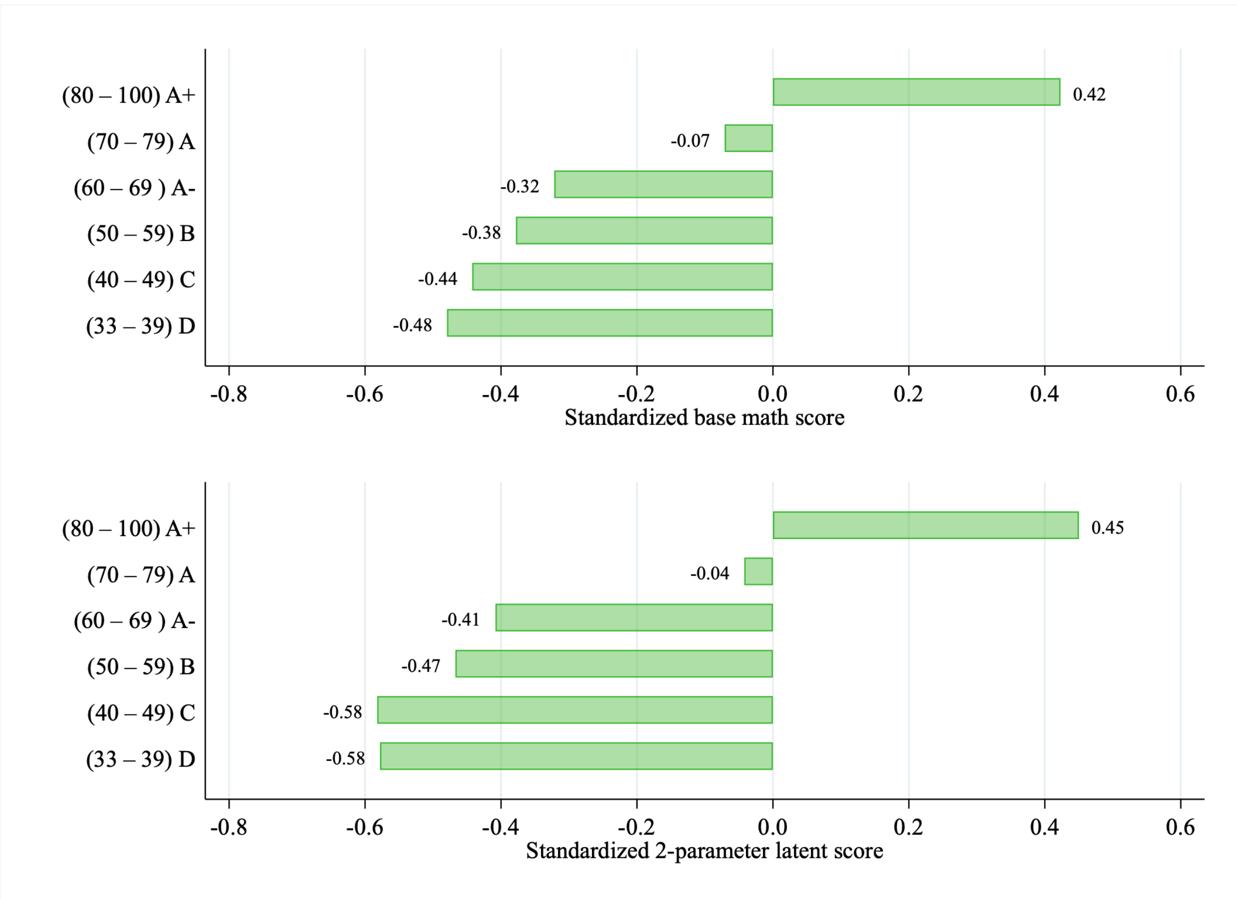


Table A.1: Study eligibility by data source

	Konnect		**SSS Sample**		**RDD Sample**		**Total**	
	N	%	N	%	N	%	N	%
Attempted	14678		12569		11720		38967	
Answered	10563	72%	8573	68%	6772	58%	25908	66%
Children in grades 6-10	5681	54%	5528	64%	2163	32%	13372	52%
Smartphones in household	3962	70%	3152	57%	1321	61%	8435	63%
Eligible and consented	3653	92%	2983	95%	1240	94%	7876	93%
Completed Baseline	3506	96%	2896	97%	1174	95%	7576	96%
Baseline / Attempted		24%		23%		10%		19%

Table A.2: Characteristics of households with children in grades 6–10, baseline and MICS-2019 samples

Variable	All	Baseline Low SES	High SES	MICS-2019 Mean
Children age 5–17	1.93	1.95	1.91	1.87
Rooms for sleeping	2.68	2.33	3.00	2.39
Flush toilet	0.58	0.20	0.93	0.51
Has mobile	1.00	1.00	1.00	0.97
Less primary, mother	0.23	0.34	0.13	0.43
Primary graduate, mother	0.38	0.44	0.32	0.43
Secondary graduate, mother	0.20	0.15	0.25	0.07
Post-secondary graduate, mother	0.19	0.08	0.31	0.06
Less primary, father	0.27	0.40	0.15	0.50
Primary graduate, father	0.27	0.33	0.22	0.30
Secondary graduate, father	0.18	0.15	0.21	0.08
Post-secondary graduate, father	0.27	0.12	0.43	0.12
		7576		20120

Table A.3: Response rates by treatment assignment

	(1)	(2)	(3)
	Round 1	Round 2	R2 Learning assessment
Edtech info	0.015	-0.046*	-0.058***
	(0.020)	(0.024)	(0.021)
Data	-0.010	-0.080**	-0.070**
	(0.032)	(0.034)	(0.030)
Teacher support	-0.010	-0.048**	-0.014
	(0.023)	(0.024)	(0.021)
Gen. info	0.020	-0.095***	-0.096***
	(0.021)	(0.032)	(0.028)
Gen. and Edtech info	0.005	-0.059**	-0.068***
	(0.020)	(0.024)	(0.021)
Edtech info X data	0.035	0.054	0.073**
	(0.034)	(0.036)	(0.032)
Gen. info X Edtech info X data	-0.042	0.051	0.056*
	(0.030)	(0.034)	(0.030)
Teacher support X data	-0.083*	0.065	0.067*
	(0.044)	(0.042)	(0.038)
Observations	8397	6981	6981
Response rate, control	0.68	0.67	0.51
P-val, joint significance	0.1989	0.0154	0.0000

Notes: Child-level data includes all respondents contacted at Round 1 and Round 2 surveys, respectively. All regressions include stratification-cell fixed effects. Robust standard errors are shown in parentheses and clustered at the household level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.4: Balance tests by pooled treatment assignment, Round 1 respondents only

	(1) All	(2) Control	(3) Edtech info	(4) Data	(5) Teacher	(6) Joint tests, all, p-val
HH size	1.89	1.88	1.95	1.87	1.90	0.730
	(0.97)	(0.95)	(0.96)	(0.96)	(1.01)	
Num. secondary children	1.27	1.24	1.32**	1.24	1.29	0.082*
	(0.51)	(0.45)	(0.57)	(0.48)	(0.61)	
Has cable/satellite TV	0.65	0.66	0.62*	0.66	0.67	0.007***
	(0.48)	(0.47)	(0.49)	(0.47)	(0.47)	
Mother present	0.49	0.50	0.50	0.51	0.49	0.631
	(0.50)	(0.50)	(0.50)	(0.50)	(0.50)	
Father present	0.50	0.50	0.48	0.48	0.51	0.603
	(0.50)	(0.50)	(0.50)	(0.50)	(0.50)	
Mother primary	0.35	0.36	0.35	0.34	0.35	0.023**
	(0.48)	(0.48)	(0.48)	(0.48)	(0.48)	
Mother secondary	0.19	0.18	0.19	0.19	0.17	0.933
	(0.39)	(0.39)	(0.39)	(0.40)	(0.38)	
Mother post-secondary	0.19	0.19	0.16	0.19	0.19	0.552
	(0.39)	(0.40)	(0.37)	(0.39)	(0.40)	
Father primary	0.25	0.25	0.27	0.26	0.25	0.803
	(0.44)	(0.43)	(0.44)	(0.44)	(0.43)	
Father secondary	0.18	0.17	0.17	0.18	0.20	0.316
	(0.38)	(0.38)	(0.37)	(0.38)	(0.40)	
Father post-secondary	0.26	0.26	0.24	0.26	0.23	0.265
	(0.44)	(0.44)	(0.42)	(0.44)	(0.42)	
Mother income	5069	5034	4468	5953	3668	0.011**
	(25644)	(26069)	(22433)	(28009)	(22706)	
Father income	53751	52451	56867	54011	51200	0.783
	(137335)	(134796)	(143636)	(139155)	(124421)	
School days/week, curr.	5.75	5.80	5.69	5.73	5.68	0.935
	(2.20)	(2.13)	(2.30)	(2.21)	(2.31)	
School days/week, Apr. 20	5.43	5.47	5.51	5.40	5.45	0.914
	(2.13)	(2.11)	(2.03)	(2.17)	(2.14)	
Has private tutor	0.60	0.59	0.61	0.60	0.61	0.981
	(0.49)	(0.49)	(0.49)	(0.49)	(0.49)	
Working for pay	0.03	0.03	0.03	0.03	0.02	0.660
	(0.17)	(0.17)	(0.17)	(0.17)	(0.15)	
Number of students	5736	1411	754	1662	587	
Number of households	5021	1249	643	1470	514	
Joint test, p-val			0.824	0.991	0.728	

Notes: Sample includes all randomized baseline respondents at the child level. Stars in columns 3–5 indicate statistically significant differences relative to the control group (column 2). Column 6 reports p-values based on F-tests of the joint significance of all eight treatment indicators, excluding respondents with missing values. P-values in the bottom row are from seemingly unrelated regressions that predict treatment assignment relative to the control group, with missing variable flags included. Standard errors are clustered at the household level. Stratification-cell fixed effects are included in all regressions. ***, **, and * indicate significance at the 1, 5, and 10 percent significance levels.

Table A.5: Balance tests by pooled treatment assignment, Round 2 respondents only

	(1) All	(2) Control	(3) Edtech info	(4) Data	(5) Teacher	(6) Joint tests, all, p-val
HH size	1.89	1.93	1.87	1.85**	1.88	0.241
	(0.97)	(1.00)	(0.85)	(0.94)	(1.02)	
Num. secondary children	1.27	1.27	1.28	1.26	1.28	0.597
	(0.50)	(0.47)	(0.49)	(0.48)	(0.60)	
Has cable/satellite TV	0.67	0.67	0.67	0.67	0.66	0.612
	(0.47)	(0.47)	(0.47)	(0.47)	(0.47)	
Mother present	0.51	0.51	0.50	0.51	0.52	0.851
	(0.50)	(0.50)	(0.50)	(0.50)	(0.50)	
Father present	0.49	0.48	0.49	0.49	0.47	0.855
	(0.50)	(0.50)	(0.50)	(0.50)	(0.50)	
Mother primary	0.34	0.36	0.35	0.33	0.32	0.416
	(0.47)	(0.48)	(0.48)	(0.47)	(0.47)	
Mother secondary	0.19	0.19	0.19	0.20	0.19	0.588
	(0.39)	(0.39)	(0.39)	(0.40)	(0.39)	
Mother post-secondary	0.21	0.21	0.21	0.22	0.21	0.912
	(0.41)	(0.41)	(0.41)	(0.41)	(0.41)	
Father primary	0.24	0.24	0.24	0.24	0.22	0.981
	(0.43)	(0.43)	(0.43)	(0.43)	(0.42)	
Father secondary	0.17	0.17	0.16	0.19	0.17	0.060*
	(0.38)	(0.37)	(0.36)	(0.39)	(0.38)	
Father post-secondary	0.28	0.27	0.29	0.27	0.27	0.664
	(0.45)	(0.45)	(0.45)	(0.44)	(0.44)	
Mother income	5100	4650	8538*	5134	3545	0.045**
	(25007)	(24178)	(34375)	(23805)	(21876)	
Father income	50855	50545	62174	47181	50812	0.115
	(130451)	(129612)	(151164)	(125999)	(126238)	
School days/week, curr.	5.78	5.86	5.68	5.75	5.72	0.265
	(2.19)	(2.09)	(2.35)	(2.19)	(2.26)	
School days/week, Apr. 20	5.49	5.50	5.61	5.44	5.57	0.852
	(2.11)	(2.10)	(2.08)	(2.15)	(2.08)	
Has private tutor	0.61	0.60	0.62	0.61	0.63	0.866
	(0.49)	(0.49)	(0.49)	(0.49)	(0.48)	
Working for pay	0.03	0.03	0.03	0.02	0.03	0.980
	(0.16)	(0.17)	(0.16)	(0.15)	(0.16)	
Number of students	3881	1161	362	1323	492	
Number of households	3375	1009	313	1155	433	
Joint test, p-val			0.372	0.278	0.673	

Notes: Sample includes all randomized baseline respondents at the child level. Stars in columns 3–5 indicate statistically significant differences relative to the control group (column 2). Column 6 reports p-values based on F-tests of the joint significance of all eight treatment indicators, excluding respondents with missing values. P-values in the bottom row are from seemingly unrelated regressions that predict treatment assignment relative to the control group, with missing variable flags included. Standard errors are clustered at the household level. Stratification-cell fixed effects are included in all regressions. ***, **, and * indicate significance at the 1, 5, and 10 percent significance levels.

Table A.6: Balance tests by pooled treatment assignment, learning assessment respondents only

	(1) All	(2) Control	(3) Edtech info	(4) Data	(5) Teacher	(6) Joint tests, all, p-val
HH size	1.79	1.83	1.79	1.74***	1.77	0.101
	(0.91)	(0.95)	(0.85)	(0.88)	(0.90)	
Num. secondary children	1.15	1.15	1.17	1.14	1.14	0.465
	(0.38)	(0.37)	(0.40)	(0.37)	(0.40)	
Has cable/satellite TV	0.67	0.67	0.68	0.66	0.66	0.529
	(0.47)	(0.47)	(0.47)	(0.47)	(0.47)	
Mother present	0.51	0.51	0.49	0.50	0.53	0.970
	(0.50)	(0.50)	(0.50)	(0.50)	(0.50)	
Father present	0.49	0.48	0.50	0.49	0.47	0.965
	(0.50)	(0.50)	(0.50)	(0.50)	(0.50)	
Mother primary	0.34	0.36	0.34	0.33	0.32	0.487
	(0.47)	(0.48)	(0.47)	(0.47)	(0.47)	
Mother secondary	0.19	0.19	0.20	0.20	0.18	0.670
	(0.39)	(0.39)	(0.40)	(0.40)	(0.38)	
Mother post-secondary	0.21	0.21	0.21	0.21	0.21	0.869
	(0.41)	(0.41)	(0.41)	(0.41)	(0.41)	
Father primary	0.24	0.24	0.24	0.24	0.23	0.970
	(0.43)	(0.43)	(0.43)	(0.43)	(0.42)	
Father secondary	0.17	0.16	0.15	0.19	0.18	0.080*
	(0.38)	(0.37)	(0.36)	(0.39)	(0.38)	
Father post-secondary	0.28	0.27	0.29	0.27	0.25	0.319
	(0.45)	(0.45)	(0.46)	(0.45)	(0.43)	
Mother income	4934	4515	6710	5406	3021	0.060*
	(24460)	(23778)	(29697)	(25016)	(19522)	
Father income	49471	49286	51682	48133	48891	0.308
	(128879)	(128434)	(132397)	(128555)	(126192)	
School days/week, curr.	5.80	5.91	5.61*	5.79	5.71	0.097*
	(2.16)	(2.02)	(2.41)	(2.15)	(2.26)	
School days/week, Apr. 20	5.51	5.51	5.63	5.44	5.55	0.906
	(2.10)	(2.10)	(2.11)	(2.14)	(2.08)	
Has private tutor	0.62	0.61	0.63	0.62	0.62	0.934
	(0.49)	(0.49)	(0.48)	(0.49)	(0.48)	
Working for pay	0.03	0.03	0.03	0.02	0.03	0.936
	(0.17)	(0.17)	(0.17)	(0.15)	(0.18)	
Number of students	3434	1031	320	1173	442	
Number of households	3218	970	301	1099	418	
Joint test, p-val			0.330	0.155	0.456	

Notes: Sample includes all randomized baseline respondents at the child level. Stars in columns 3–5 indicate statistically significant differences relative to the control group (column 2). Column 6 reports p-values based on F-tests of the joint significance of all eight treatment indicators, excluding respondents with missing values. P-values in the bottom row are from seemingly unrelated regressions that predict treatment assignment relative to the control group, with missing variable flags included. Standard errors are clustered at the household level. Stratification-cell fixed effects are included in all regressions. ***, **, and * indicate significance at the 1, 5, and 10 percent significance levels.

Table A.7: Impact of interventions on tech-based resource use, disaggregated, all treatment indicators

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. All						
	Sangsad TV	Video lessons	Robi platform	Teacher remotely	Remote classes	Tech index
Edtech info	-0.049***	-0.029	-0.006	0.009	-0.008	-0.046*
	(0.017)	(0.019)	(0.009)	(0.015)	(0.011)	(0.024)
Data	-0.008	0.098***	-0.015	-0.004	-0.003	0.042
	(0.030)	(0.037)	(0.013)	(0.024)	(0.019)	(0.042)
Teacher support	-0.011	0.005	-0.010	0.021	0.007	0.010
	(0.021)	(0.023)	(0.010)	(0.019)	(0.015)	(0.032)
Gen. info	0.039*	0.015	0.001	0.022	0.009	0.043
	(0.020)	(0.021)	(0.010)	(0.017)	(0.013)	(0.027)
Gen. and Edtech info	0.008	-0.012	-0.004	0.001	0.031**	0.014
	(0.019)	(0.021)	(0.010)	(0.015)	(0.014)	(0.027)
Edtech info X data	-0.017	-0.128***	0.035**	0.010	0.020	-0.042
	(0.031)	(0.038)	(0.015)	(0.025)	(0.021)	(0.046)
Gen. info X Edtech info X data	0.042	0.059*	-0.018	0.011	-0.036*	0.010
	(0.028)	(0.031)	(0.015)	(0.023)	(0.020)	(0.042)
Teacher support X data	-0.027	-0.037	0.007	-0.011	-0.016	-0.073
	(0.041)	(0.047)	(0.021)	(0.035)	(0.025)	(0.053)
DV mean, control	0.20	0.25	0.05	0.12	0.08	-0.00
Observations	5715	5715	5715	5715	5715	5715
Panel B. Low-SES Households						
	Sangsad TV	Video lessons	Robi platform	Teacher remotely	Remote classes	Tech index
Edtech info	-0.049**	-0.027	-0.004	0.018	-0.006	-0.040
	(0.023)	(0.026)	(0.010)	(0.020)	(0.012)	(0.030)
Data	-0.016	0.129**	0.007	-0.009	0.002	0.038
	(0.043)	(0.050)	(0.019)	(0.028)	(0.020)	(0.053)
Teacher support	-0.040	-0.025	-0.006	0.024	0.022	0.002
	(0.028)	(0.030)	(0.013)	(0.025)	(0.017)	(0.044)
Gen. info	-0.036	-0.019	-0.007	0.020	0.018	0.000
	(0.025)	(0.027)	(0.009)	(0.020)	(0.014)	(0.029)
Gen. and Edtech info	0.012	-0.006	-0.006	-0.011	0.033**	-0.001
	(0.027)	(0.027)	(0.009)	(0.017)	(0.015)	(0.030)
Edtech info X data	-0.034	-0.193***	-0.009	0.029	0.014	-0.083
	(0.044)	(0.051)	(0.019)	(0.030)	(0.022)	(0.055)
Gen. info X Edtech info X data	-0.001	0.028	-0.005	0.001	-0.045**	-0.000
	(0.037)	(0.040)	(0.013)	(0.027)	(0.022)	(0.043)
Teacher support X data	-0.001	0.007	0.028	0.009	-0.026	0.019
	(0.054)	(0.069)	(0.029)	(0.049)	(0.025)	(0.072)
DV mean, control	0.18	0.20	0.02	0.08	0.04	-0.10
Observations	2881	2881	2881	2881	2881	2881
Panel C. High-SES Households						
	Sangsad TV	Video lessons	Robi platform	Teacher remotely	Remote classes	Tech index
Edtech info	-0.058**	-0.028	-0.004	-0.003	-0.017	-0.066*
	(0.026)	(0.029)	(0.016)	(0.023)	(0.020)	(0.038)
Data	-0.000	0.071	-0.036*	-0.002	0.003	0.044
	(0.042)	(0.054)	(0.018)	(0.039)	(0.031)	(0.065)
Teacher support	0.014	0.036	-0.016	0.016	-0.012	0.009
	(0.033)	(0.036)	(0.016)	(0.029)	(0.025)	(0.044)
Gen. info	0.128***	0.061*	0.011	0.023	0.005	0.102**
	(0.033)	(0.034)	(0.020)	(0.028)	(0.024)	(0.048)
Gen. and Edtech info	0.002	-0.017	0.001	0.017	0.031	0.031
	(0.028)	(0.032)	(0.018)	(0.025)	(0.023)	(0.045)
Edtech info X data	-0.004	-0.055	0.080***	-0.001	0.013	0.004
	(0.044)	(0.056)	(0.024)	(0.042)	(0.035)	(0.074)
Gen. info X Edtech info X data	0.084**	0.084*	-0.038	0.018	-0.036	0.009
	(0.043)	(0.049)	(0.029)	(0.038)	(0.035)	(0.071)
Teacher support X data	-0.041	-0.085	0.003	-0.012	0.004	-0.121
	(0.059)	(0.067)	(0.031)	(0.053)	(0.044)	(0.077)
DV mean, control	0.22	0.31	0.07	0.15	0.12	0.10
Observations	2834	2834	2834	2834	2834	2834
H vs. L: info	0.930	0.895	0.975	0.430	0.643	0.609
H vs. L: data	0.901	0.495	0.063	0.814	0.862	0.990
H vs. L: teacher	0.310	0.168	0.604	0.978	0.227	0.899

Notes: Sample includes all Round 1 survey respondents. All regressions include stratification-cell fixed effects. Baseline covariates selected using post-double-selection lasso with 5-fold cross-validation. Robust standard errors are shown in parentheses and clustered at the household level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.8: Impact of interventions on non-tech-based resource use, disaggregated, all treatment indicators

	(1)	(2)	(3)	(4)
Panel A. All				
	Textbooks	Exercise books	Teacher in-person	Non-Tech index
Edtech info	-0.009	0.008	0.025	-0.001
	(0.012)	(0.022)	(0.023)	(0.027)
Data	-0.007	0.039	0.018	0.038
	(0.020)	(0.037)	(0.036)	(0.050)
Teacher support	-0.030*	-0.027	-0.079***	-0.112***
	(0.016)	(0.026)	(0.027)	(0.031)
Gen. info	-0.003	0.006	0.014	-0.008
	(0.012)	(0.024)	(0.024)	(0.028)
Gen. and Edtech info	0.005	-0.025	-0.026	-0.012
	(0.011)	(0.022)	(0.024)	(0.027)
Edtech info X data	0.006	-0.043	-0.003	-0.043
	(0.021)	(0.039)	(0.038)	(0.052)
Gen. info X Edtech info X data	-0.000	0.036	0.011	0.012
	(0.017)	(0.034)	(0.035)	(0.041)
Teacher support X data	0.031	0.014	0.116**	0.095*
	(0.026)	(0.047)	(0.049)	(0.049)
DV mean, control	0.94	0.32	0.62	-0.00
Observations	5715	5715	5715	5715
Panel B. Low-SES Households				
	Textbooks	Exercise books	Teacher in-person	Non-Tech index
Edtech info	-0.009	-0.008	0.030	-0.001
	(0.017)	(0.033)	(0.033)	(0.038)
Data	-0.027	-0.057	0.008	-0.092
	(0.035)	(0.053)	(0.057)	(0.062)
Teacher support	-0.010	-0.024	-0.103***	-0.075*
	(0.022)	(0.036)	(0.037)	(0.043)
Gen. info	-0.003	0.004	-0.007	-0.009
	(0.018)	(0.034)	(0.034)	(0.040)
Gen. and Edtech info	0.011	-0.008	-0.022	-0.012
	(0.017)	(0.032)	(0.033)	(0.038)
Edtech info X data	0.029	0.068	0.037	0.117*
	(0.036)	(0.055)	(0.059)	(0.065)
Gen. info X Edtech info X data	-0.022	0.001	-0.008	-0.021
	(0.027)	(0.049)	(0.050)	(0.058)
Teacher support X data	0.047*	-0.045	0.000	-0.005
	(0.027)	(0.069)	(0.073)	(0.065)
DV mean, control	0.93	0.32	0.60	-0.03
Observations	2881	2881	2881	2881
Panel C. High-SES Households				
	Textbooks	Exercise books	Teacher in-person	Non-Tech index
Edtech info	-0.009	0.022	0.008	-0.007
	(0.017)	(0.033)	(0.033)	(0.040)
Data	0.019	0.115**	0.036	0.170**
	(0.021)	(0.054)	(0.047)	(0.072)
Teacher support	-0.043*	-0.035	-0.059	-0.144***
	(0.024)	(0.038)	(0.039)	(0.046)
Gen. info	0.001	-0.001	0.030	-0.001
	(0.017)	(0.036)	(0.035)	(0.040)
Gen. and Edtech info	0.001	-0.045	-0.020	-0.014
	(0.015)	(0.032)	(0.033)	(0.038)
Edtech info X data	-0.021	-0.135**	-0.056	-0.205***
	(0.023)	(0.057)	(0.051)	(0.075)
Gen. info X Edtech info X data	0.012	0.084*	0.014	0.039
	(0.022)	(0.049)	(0.051)	(0.058)
Teacher support X data	0.017	0.067	0.217***	0.183**
	(0.042)	(0.066)	(0.064)	(0.073)
DV mean, control	0.95	0.33	0.64	0.03
Observations	2834	2834	2834	2834
H vs. L: info	0.865	0.621	0.572	0.979
H vs. L: data	0.222	0.019	0.756	0.006
H vs. L: teacher	0.302	0.899	0.490	0.311

Notes: Sample includes all Round 1 survey respondents. All regressions include stratification-cell fixed effects. Baseline covariates selected using post-double-selection lasso with 5-fold cross-validation. Robust standard errors are shown in parentheses and clustered at the household level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.9: Impact of interventions on parental investment, all treatment indicators

	(1)	(2)	(3)	(4)
Panel A. All				
	Hours parent helped	Private tutoring	Money on tutoring	Money on other education
Edtech info	-0.492	0.046**	163.167**	-26.870**
	(0.347)	(0.021)	(72.106)	(10.664)
Data	0.249	0.080**	149.976	18.118
	(0.716)	(0.032)	(101.620)	(18.128)
Teacher support	0.008	-0.047*	52.067	7.007
	(0.422)	(0.024)	(85.859)	(14.451)
Gen. info	0.090	0.004	-7.480	4.600
	(0.400)	(0.023)	(71.147)	(13.326)
Gen. and Edtech info	0.038	-0.028	24.877	-8.368
	(0.379)	(0.022)	(65.013)	(12.246)
Edtech info X data	-0.508	-0.036	-78.588	-30.184
	(0.735)	(0.034)	(107.125)	(18.529)
Gen. info X Edtech info X data	0.866	-0.017	-93.054	26.784
	(0.568)	(0.032)	(98.144)	(17.789)
Teacher support X data	-0.742	0.084*	176.144	-33.959
	(0.822)	(0.044)	(150.463)	(24.659)
DV mean, control	6.57	0.64	1027.82	138.56
Observations	5359	5688	5359	5065
Panel B. Low-SES Households				
	Hours parent helped	Private tutoring	Money on tutoring	Money on other education
Edtech info	-1.008**	0.016	137.198**	-31.108***
	(0.442)	(0.031)	(68.635)	(10.746)
Data	-0.084	0.052	188.911*	40.706*
	(1.102)	(0.051)	(112.416)	(24.572)
Teacher support	0.073	-0.064*	39.332	-9.654
	(0.543)	(0.035)	(70.636)	(14.508)
Gen. info	-0.031	-0.017	-7.690	-19.378
	(0.522)	(0.032)	(58.865)	(12.977)
Gen. and Edtech info	0.129	-0.040	108.442*	-9.458
	(0.494)	(0.032)	(60.328)	(13.367)
Edtech info X data	-0.013	0.015	-65.065	-65.294***
	(1.132)	(0.053)	(119.805)	(24.179)
Gen. info X Edtech info X data	0.456	-0.019	-161.454*	14.078
	(0.763)	(0.048)	(95.130)	(18.352)
Teacher support X data	-0.898	0.026	13.951	-14.264
	(1.123)	(0.064)	(117.462)	(27.375)
DV mean, control	5.81	0.59	583.89	78.28
Observations	2698	2866	2735	2542
Panel C. High-SES Households				
	Hours parent helped	Private tutoring	Money on tutoring	Money on other education
Edtech info	0.049	0.062**	151.463	-20.203
	(0.553)	(0.030)	(134.601)	(18.993)
Data	0.657	0.089**	43.893	0.342
	(0.962)	(0.042)	(180.804)	(27.051)
Teacher support	-0.022	-0.009	53.049	21.554
	(0.664)	(0.035)	(163.535)	(25.738)
Gen. info	0.464	0.011	-20.272	27.787
	(0.625)	(0.034)	(143.650)	(24.675)
Gen. and Edtech info	0.016	-0.011	-78.227	-4.352
	(0.601)	(0.031)	(118.346)	(20.839)
Edtech info X data	-0.942	-0.071	-103.642	7.186
	(0.983)	(0.045)	(189.529)	(28.836)
Gen. info X Edtech info X data	1.485*	-0.022	59.173	32.283
	(0.873)	(0.045)	(179.146)	(31.388)
Teacher support X data	-0.666	0.122**	276.502	-52.455
	(1.251)	(0.062)	(276.740)	(41.855)
DV mean, control	7.31	0.68	1474.41	195.61
Observations	2661	2822	2624	2522
H vs. L: info	0.100	0.361	0.596	0.556
H vs. L: data	0.665	0.481	0.761	0.257
H vs. L: teacher	0.941	0.416	0.844	0.207

Notes: All expenses reported in taka. Sample includes all Round 1 survey respondents. All regressions include stratification-cell fixed effects. Baseline covariates selected using post-double-selection lasso with 5-fold cross-validation. Robust standard errors are shown in parentheses and clustered at the household level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.10: Impact of interventions on student learning (math), endline, all treatment indicators

	(1)	(2)
Panel A. All		
	Unadjusted score	IRT, 2pl
Edtech info	0.149**	0.150**
	(0.059)	(0.058)
Data	-0.075	0.048
	(0.087)	(0.083)
Teacher support	0.001	-0.016
	(0.059)	(0.058)
Gen. info	-0.205**	-0.120
	(0.084)	(0.086)
Gen. and Edtech info	0.008	0.047
	(0.061)	(0.061)
Edtech info X data	0.080	-0.019
	(0.092)	(0.088)
Gen. info X Edtech info X data	0.041	-0.004
	(0.084)	(0.084)
Teacher support X data	-0.055	-0.081
	(0.105)	(0.102)
DV mean, control	0.01	0.00
Observations	3433	3433
Panel B. Low-SES Households		
	Unadjusted score	IRT, 2pl
Edtech info	0.013	0.084
	(0.099)	(0.097)
Data	-0.213	-0.153
	(0.141)	(0.132)
Teacher support	0.014	0.057
	(0.097)	(0.090)
Gen. info	-0.302**	-0.258*
	(0.144)	(0.142)
Gen. and Edtech info	0.112	0.152
	(0.097)	(0.095)
Edtech info X data	0.251*	0.214
	(0.146)	(0.137)
Gen. info X Edtech info X data	-0.123	-0.168
	(0.132)	(0.131)
Teacher support X data	-0.036	-0.107
	(0.161)	(0.143)
DV mean, control	-0.15	-0.20
Observations	1615	1615
Panel C. High-SES Households		
	Unadjusted score	IRT, 2pl
Edtech info	0.238***	0.213***
	(0.078)	(0.075)
Data	0.028	0.202*
	(0.116)	(0.110)
Teacher support	-0.032	-0.102
	(0.077)	(0.081)
Gen. info	-0.117	-0.006
	(0.100)	(0.105)
Gen. and Edtech info	-0.072	0.001
	(0.081)	(0.080)
Edtech info X data	-0.070	-0.194
	(0.126)	(0.119)
Gen. info X Edtech info X data	0.199*	0.121
	(0.115)	(0.115)
Teacher support X data	-0.070	-0.069
	(0.146)	(0.146)
DV mean, control	0.15	0.18
Observations	1808	1808
H vs. L: info	0.115	0.344
H vs. L: data	0.169	0.029
H vs. L: teacher	0.913	0.388

Notes: Standardized score includes sum of scores on 4 math questions, normalized to the grade-specific control group. IRT adjusted score shows predicted latent ability from full set of math questions, normalized to control group mean (not grade-specific). Sample includes all Round 2 survey respondents. All regressions include stratification-cell fixed effects. Baseline covariates selected using post-double-selection lasso with 5-fold cross-validation. Robust standard errors are shown in parentheses and clustered at the household level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.11: Impact of interventions on tech-based resource use, disaggregated, stratification cell FE only

	(1)	(2)	(3)	(4)	(5)	(6)
	Panel A. All					
	Sangsad TV	Video lessons	Robi platform	Teacher remotely	Remote classes	Tech index
Edtech info	-0.059***	-0.036*	-0.009	0.005	-0.013	-0.057**
	(0.018)	(0.020)	(0.009)	(0.015)	(0.012)	(0.025)
Data	0.011	0.094***	-0.010	-0.003	-0.003	0.032
	(0.031)	(0.037)	(0.014)	(0.024)	(0.021)	(0.043)
Teacher support	-0.011	-0.001	-0.016	0.014	-0.002	-0.004
	(0.022)	(0.024)	(0.011)	(0.019)	(0.015)	(0.032)
DV mean, control	0.20	0.25	0.05	0.12	0.08	-0.00
Observations	5715	5715	5715	5715	5715	5715
	Panel B. Low-SES Households					
	Sangsad TV	Video lessons	Robi platform	Teacher remotely	Remote classes	Tech index
Edtech info	-0.059**	-0.035	-0.005	0.009	-0.013	-0.054*
	(0.024)	(0.027)	(0.010)	(0.020)	(0.013)	(0.030)
Data	0.002	0.116**	0.008	-0.012	-0.011	0.028
	(0.045)	(0.050)	(0.019)	(0.028)	(0.020)	(0.053)
Teacher support	-0.028	-0.031	-0.007	0.018	0.015	-0.011
	(0.031)	(0.031)	(0.013)	(0.025)	(0.018)	(0.045)
DV mean, control	0.18	0.20	0.02	0.08	0.04	-0.10
Observations	2787	2787	2787	2787	2787	2787
	Panel C. High-SES Households					
	Sangsad TV	Video lessons	Robi platform	Teacher remotely	Remote classes	Tech index
Edtech info	-0.064**	-0.029	-0.009	-0.004	-0.015	-0.059
	(0.026)	(0.030)	(0.015)	(0.023)	(0.020)	(0.039)
Data	0.009	0.066	-0.028	0.001	0.008	0.033
	(0.044)	(0.053)	(0.023)	(0.039)	(0.035)	(0.066)
Teacher support	-0.005	0.021	-0.031*	0.005	-0.025	-0.015
	(0.033)	(0.036)	(0.017)	(0.028)	(0.025)	(0.046)
DV mean, control	0.22	0.30	0.07	0.15	0.11	0.09
Observations	2928	2928	2928	2928	2928	2928
H vs. L: info	0.879	0.868	0.769	0.774	0.956	0.960
H vs. L: data	0.826	0.441	0.153	0.639	0.713	0.906
H vs. L: teacher	0.484	0.200	0.336	0.936	0.220	0.827

Notes: Sample includes all Round 1 survey respondents. All regressions include stratification-cell fixed effects and indicators for general information, general and app information, and treatment interactions. Robust standard errors are shown in parentheses and clustered at the household level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.12: Impact of interventions on non-tech-based resource use, disaggregated, stratification cell FE only

	(1)	(2)	(3)	(4)
Panel A. All				
	Textbooks	Exercise books	Teacher in-person	Non-Tech index
Edtech info	-0.011	0.004	0.025	-0.003
	(0.013)	(0.023)	(0.023)	(0.028)
Data	-0.007	0.036	0.033	0.046
	(0.020)	(0.038)	(0.037)	(0.051)
Teacher support	-0.033**	-0.038	-0.084***	-0.120***
	(0.017)	(0.026)	(0.028)	(0.032)
DV mean, control	0.94	0.32	0.62	-0.00
Observations	5715	5715	5715	5715
Panel B. Low-SES Households				
	Textbooks	Exercise books	Teacher in-person	Non-Tech index
Edtech info	-0.004	-0.023	0.030	-0.003
	(0.018)	(0.033)	(0.035)	(0.040)
Data	-0.034	-0.059	0.030	-0.083
	(0.035)	(0.051)	(0.059)	(0.064)
Teacher support	-0.011	-0.030	-0.094**	-0.090**
	(0.023)	(0.037)	(0.040)	(0.045)
DV mean, control	0.93	0.32	0.60	-0.03
Observations	2787	2787	2787	2787
Panel C. High-SES Households				
	Textbooks	Exercise books	Teacher in-person	Non-Tech index
Edtech info	-0.016	0.029	0.011	-0.008
	(0.017)	(0.033)	(0.033)	(0.039)
Data	0.020	0.126**	0.062	0.177**
	(0.021)	(0.054)	(0.047)	(0.072)
Teacher support	-0.048**	-0.039	-0.063	-0.139***
	(0.024)	(0.037)	(0.039)	(0.045)
DV mean, control	0.95	0.33	0.63	0.02
Observations	2928	2928	2928	2928
H vs. L: info	0.796	0.298	0.611	0.965
H vs. L: data	0.208	0.008	0.784	0.008
H vs. L: teacher	0.267	0.953	0.792	0.458

Notes: Sample includes all Round 1 survey respondents. All regressions include stratification-cell fixed effects and indicators for general information, general and app information, and treatment interactions. Robust standard errors are shown in parentheses and clustered at the household level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.13: Impact of interventions on parental investment, stratification cell FE only

	(1)	(2)	(3)	(4)
		Panel A. All		
	Hours parent helped	Private tutoring	Money on tutoring	Money on other education
Edtech info	-0.601*	0.042*	169.919**	-33.544***
	(0.356)	(0.023)	(86.476)	(11.483)
Data	0.201	0.107***	108.895	15.213
	(0.729)	(0.035)	(115.880)	(19.887)
Teacher support	-0.051	-0.051*	-6.038	0.161
	(0.441)	(0.027)	(98.056)	(15.335)
DV mean, control	6.57	0.64	1027.82	138.56
Observations	5359	5688	5359	5065
		Panel B. Low-SES Households		
	Hours parent helped	Private tutoring	Money on tutoring	Money on other education
Edtech info	-1.131**	0.014	90.759	-35.586***
	(0.450)	(0.035)	(77.597)	(11.052)
Data	-0.282	0.077	126.159	35.903
	(1.080)	(0.058)	(113.390)	(24.805)
Teacher support	0.158	-0.060	-11.292	-11.873
	(0.604)	(0.040)	(79.505)	(15.035)
DV mean, control	5.82	0.59	589.29	80.12
Observations	2613	2772	2643	2458
		Panel C. High-SES Households		
	Hours parent helped	Private tutoring	Money on tutoring	Money on other education
Edtech info	-0.001	0.062**	273.766*	-22.186
	(0.543)	(0.030)	(152.322)	(19.470)
Data	0.627	0.127***	36.126	-1.264
	(0.989)	(0.045)	(197.516)	(30.920)
Teacher support	-0.112	-0.031	-34.254	8.063
	(0.651)	(0.037)	(177.980)	(26.435)
DV mean, control	7.25	0.68	1438.55	190.40
Observations	2746	2916	2716	2606
H vs. L: info	0.113	0.361	0.198	0.455
H vs. L: data	0.640	0.560	0.634	0.288
H vs. L: teacher	0.751	0.770	0.854	0.328

Notes: All expenses reported in taka. Sample includes all Round 1 survey respondents . All regressions include stratification-cell fixed effects and indicators for general information, general and app information, and treatment interactions. Robust standard errors are shown in parentheses and clustered at the household level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.14: Impact of interventions on student learning (math), endline, stratification cell FE only

	(1)	(2)
Panel A. All		
	Unadjusted score	IRT, 2pl
Edtech info	0.113*	0.123**
	(0.063)	(0.062)
Data	-0.035	0.039
	(0.096)	(0.092)
Teacher support	-0.016	-0.037
	(0.064)	(0.062)
DV mean, control	0.01	0.00
Observations	3433	3433
Panel B. Low-SES Households		
	Unadjusted score	IRT, 2pl
Edtech info	0.015	0.066
	(0.101)	(0.097)
Data	-0.241	-0.204
	(0.150)	(0.143)
Teacher support	0.009	0.045
	(0.099)	(0.092)
DV mean, control	-0.15	-0.21
Observations	1561	1561
Panel C. High-SES Households		
	Unadjusted score	IRT, 2pl
Edtech info	0.206***	0.190**
	(0.080)	(0.079)
Data	0.109	0.220*
	(0.121)	(0.114)
Teacher support	-0.053	-0.141
	(0.085)	(0.087)
DV mean, control	0.15	0.17
Observations	1862	1862
H vs. L: info	0.094	0.214
H vs. L: data	0.083	0.018
H vs. L: teacher	0.937	0.284

Notes: Standardized score includes sum of scores on 4 math questions, normalized to the grade-specific control group. IRT adjusted score shows predicted latent ability from full set of math questions, normalized to control group mean (not grade-specific). Sample includes all Round 2 survey respondents. All regressions include stratification-cell fixed effects and indicators for general information, general and app information, and treatment interactions. Robust standard errors are shown in parentheses and clustered at the household level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.15: Impact of interventions on student time investment, second follow-up

	Follow-up 1		Follow-up 2	
	(1)	(2)	(3)	(4)
	Days/week schoolwork	Hrs/week schoolwork	Days/week schoolwork	Hrs/week schoolwork
Edtech info	0.067	0.370	-0.078	-1.070
	(0.111)	(0.715)	(0.140)	(0.716)
Data	0.082	0.431	-0.483**	-1.532
	(0.171)	(1.153)	(0.232)	(1.057)
Teacher support	0.065	-0.341	-0.082	-0.693
	(0.133)	(0.827)	(0.149)	(0.700)
DV mean, control	5.65	19.03	5.46	15.35
Observations	5619	5168	4245	4194

Notes: Columns 1–2 sample includes all Round 1 survey respondents .. Columns 3–4 sample includes all Round 2 survey respondents. All regressions include stratification-cell fixed effects and indicators for general information, general and app information, and treatment interactions.. Baseline covariates selected using post-double-selection lasso with 5-fold cross-validation. Robust standard errors are shown in parentheses and clustered at the household level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.16: Impact of interventions on student engagement and motivation, endline

	(1)	(2)	(3)	(4)
	Student engagement index	Hope post-secondary	Attending in-person classes	Index (7)
App. info	0.038	-0.014	0.007	0.021
	(0.061)	(0.020)	(0.008)	(0.036)
Data	0.023	-0.022	0.005	0.014
	(0.091)	(0.029)	(0.011)	(0.057)
Teacher support	0.005	-0.006	-0.001	-0.006
	(0.062)	(0.020)	(0.006)	(0.036)
DV mean, control	-0.01	0.89	0.01	-0.00
Observations	3397	3297	3442	3442
infotest				
datatestall				
teachtestall				

Notes: Sample includes all Round 2 survey respondents . All regressions include stratification-cell fixed effects and indicators for general information, general and app information, and treatment interactions.. Baseline covariates selected using post-double-selection lasso with 5-fold cross-validation. Robust standard errors are shown in parentheses and clustered at the household level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.17: Intensive-margin impact of interventions on learning resources

	(1)	(2)	(3)	(4)	(5)	(6)
	Panel A. Tech-dependent learning resources					
	Sangsad TV	Video lessons	App platform	Teacher remotely	Remote classes	Index
App. info	-11.214**	-21.607**	-2.554	-3.694	-4.797	-0.039
	(5.508)	(8.760)	(2.703)	(4.658)	(5.840)	(0.024)
Data	10.894	18.208	-2.556	-7.776	-1.969	0.024
	(12.681)	(17.287)	(3.185)	(7.829)	(10.002)	(0.045)
Teacher support	-6.719	-6.358	-2.138	-2.073	-0.550	-0.018
	(5.435)	(11.166)	(2.389)	(5.597)	(7.435)	(0.024)
DV mean, control	36.70	74.73	6.84	18.62	25.78	0.00
Observations	5409	5321	5628	5507	5621	5715
infotest						
datatestall						
teachtestall						
	Panel B. Non tech-dependent learning resources					
	Textbooks	Exercise books	Teacher in-person	Index		
App. info	23.814	-15.462	21.414	0.018		
	(37.688)	(13.714)	(20.711)	(0.030)		
Data	67.219	27.035	43.191	0.162*		
	(69.574)	(28.551)	(38.035)	(0.097)		
Teacher support	5.764	-6.331	-32.114	-0.045		
	(44.011)	(19.782)	(22.892)	(0.029)		
DV mean, control	996.71	117.18	284.49	0.01		
Observations	5226	5142	5312	5715		

Notes: Sample includes all Round 1 survey respondents . All regressions include stratification-cell fixed effects and indicators for general information, general and app information, and treatment interactions.. Baseline covariates selected using post-double-selection lasso with 5-fold cross-validation. Robust standard errors are shown in parentheses and clustered at the household level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.18: Impact of interventions on phone and data use

	(1)	(2)	(3)	(4)
Panel A. Main effects				
	Smartphone use	Pre-paid data use	Pre-paid GB used	Spent on phone/internet (taka)
App. info	-0.060***	-0.035*	0.137	-27.643***
	(0.021)	(0.018)	(0.892)	(10.717)
Data	0.110***	0.056*	2.853	19.640
	(0.037)	(0.033)	(2.613)	(18.282)
Teacher support	0.015	0.023	0.215	5.852
	(0.025)	(0.022)	(0.799)	(14.487)
DV mean, control	0.34	0.20	2.03	138.56
Observations	5715	5715	5321	5065
Panel B. Persistence effects				
		Smartphone use	Pre-paid data use	Pre-paid GB used
App. info		-0.014	-0.012	-1.110
		(0.026)	(0.023)	(0.731)
Data		0.038	0.029	-0.424
		(0.038)	(0.036)	(0.878)
Teacher support		-0.031	-0.034	-0.260
		(0.025)	(0.023)	(1.024)
DV mean, control		0.29	0.19	2.35
Observations		4326	4326	4039

Notes: Panel A includes Round 1 survey respondents .. Panel B includes all Round 2 survey respondents. All regressions include stratification-cell fixed effects and indicators for general information, general and app information, and treatment interactions.. Baseline covariates selected using post-double-selection lasso with 5-fold cross-validation. Robust standard errors are shown in parentheses and clustered at the household level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.19: Persistence of impact of outreach on non-tech learning resources

	(1)	(2)	(3)	(4)	(5)	(6)
	Panel A. All					
	Sangsad TV	Video lessons	App platform	Teacher remotely	Remote classes	Tech index
Edtech info	-0.019	-0.012	0.003	-0.004	0.006	-0.046*
	(0.019)	(0.023)	(0.013)	(0.018)	(0.016)	(0.024)
Data	0.047	0.049	0.009	0.032	-0.000	0.042
	(0.030)	(0.034)	(0.017)	(0.030)	(0.023)	(0.042)
Teacher support	0.017	-0.028	-0.004	-0.029*	-0.003	0.010
	(0.021)	(0.023)	(0.012)	(0.018)	(0.015)	(0.032)
DV mean, control	0.13	0.22	0.05	0.12	0.08	-0.00
Observations	4326	4326	4326	4326	4326	5715
	Panel B. Low-SES Households					
	Sangsad TV	Video lessons	App platform	Teacher remotely	Remote classes	Tech index
Edtech info	-0.017	-0.003	0.001	-0.010	-0.010	-0.040
	(0.026)	(0.030)	(0.012)	(0.022)	(0.016)	(0.030)
Data	0.024	0.044	0.018	0.035	0.028	0.038
	(0.041)	(0.045)	(0.023)	(0.039)	(0.031)	(0.053)
Teacher support	0.015	-0.023	0.011	-0.019	0.002	0.002
	(0.027)	(0.028)	(0.013)	(0.023)	(0.015)	(0.044)
DV mean, control	0.11	0.15	0.02	0.08	0.04	-0.10
Observations	2064	2064	2064	2064	2064	2881
	Panel C. High-SES Households					
	Sangsad TV	Video lessons	App platform	Teacher remotely	Remote classes	Tech index
Edtech info	-0.021	-0.032	0.003	0.004	0.030	-0.066*
	(0.029)	(0.036)	(0.022)	(0.029)	(0.029)	(0.038)
Data	0.077	0.060	-0.000	0.030	-0.024	0.044
	(0.047)	(0.053)	(0.028)	(0.045)	(0.033)	(0.065)
Teacher support	0.022	-0.027	-0.013	-0.042	-0.006	0.009
	(0.031)	(0.037)	(0.020)	(0.028)	(0.026)	(0.044)
DV mean, control	0.15	0.28	0.07	0.15	0.11	0.10
Observations	2258	2258	2258	2258	2258	2834
H vs. L: info	0.808	0.548	0.806	0.812	0.140	0.609
H vs. L: data	0.545	0.891	0.650	0.997	0.329	0.990
H vs. L: teacher	0.873	0.814	0.391	0.471	0.999	0.899

Notes: App platform is a binary indicator for whether student used targeted learning app in the past month. The tech-index is an equally weighted index of binary indicators for whether the student used each of 5 tech-based learning resources, standardized to the control group. Sample includes all Round 2 survey respondents . All regressions include stratification-cell fixed effects and indicators for general information, general and app information, and treatment interactions.. Baseline covariates selected using post-double-selection lasso with 5-fold cross-validation. Robust standard errors are shown in parentheses and clustered at the household level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.20: Persistence of impact of outreach on non-tech learning resources

	(1)	(2)	(3)	(4)
Panel A. All				
	Textbooks	Exercise books	Teacher in-person	Non-tech index
Edtech info	0.000	-0.016	-0.020	-0.018
	(0.012)	(0.028)	(0.029)	(0.033)
Data	0.001	0.029	-0.054	-0.009
	(0.018)	(0.042)	(0.040)	(0.040)
Teacher support	0.007	0.015	-0.027	-0.004
	(0.012)	(0.030)	(0.029)	(0.031)
DV mean, control	0.95	0.41	0.48	0.00
Observations	4326	4326	4326	4326
Panel B. Low-SES Households				
	Textbooks	Exercise books	Teacher in-person	Non-tech index
Edtech info	0.004	-0.023	0.030	0.035
	(0.019)	(0.042)	(0.043)	(0.048)
Data	0.009	0.053	-0.086	0.010
	(0.027)	(0.064)	(0.063)	(0.059)
Teacher support	0.036**	-0.013	-0.034	0.022
	(0.015)	(0.043)	(0.043)	(0.040)
DV mean, control	0.94	0.41	0.46	-0.02
Observations	2064	2064	2064	2064
Panel C. High-SES Households				
	Textbooks	Exercise books	Teacher in-person	Non-tech index
Edtech info	0.000	-0.010	-0.054	-0.061
	(0.016)	(0.040)	(0.041)	(0.044)
Data	-0.013	-0.004	-0.015	-0.037
	(0.024)	(0.057)	(0.056)	(0.055)
Teacher support	-0.030	0.033	-0.002	-0.033
	(0.020)	(0.043)	(0.042)	(0.049)
DV mean, control	0.95	0.41	0.49	0.02
Observations	2258	2258	2258	2258
H vs. L: info	0.582	0.842	0.184	0.124
H vs. L: data	0.528	0.458	0.572	0.464
H vs. L: teacher	0.003	0.542	0.611	0.283

Notes: The non-tech index is equally weighted index of binary indicators for whether the student used each of these three 3 non-tech-based learning resources, standardized to the control group. Sample includes all Round 2 survey respondents . All regressions include stratification-cell fixed effects and indicators for general information, general and app information, and treatment interactions.. Baseline covariates selected using post-double-selection lasso with 5-fold cross-validation. Robust standard errors are shown in parentheses and clustered at the household level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.21: Persistence of impact of outreach on parental investment

	(1)	(2)	(3)
Panel A. All			
	Hours parent helped	Private tutoring	Money on tutoring
Edtech info	0.032	0.010	29.708
	(0.337)	(0.028)	(86.071)
Data	0.076	0.049	116.569
	(0.539)	(0.039)	(119.246)
Teacher support	0.476	-0.034	19.079
	(0.364)	(0.028)	(80.044)
DV mean, control	4.56	0.48	743.05
Observations	4185	4299	4256
Panel B. Low-SES Households			
	Hours parent helped	Private tutoring	Money on tutoring
Edtech info	-0.104	0.052	-17.323
	(0.447)	(0.041)	(62.888)
Data	-0.459	-0.018	127.996
	(0.672)	(0.060)	(142.077)
Teacher support	0.237	-0.078*	-137.442**
	(0.498)	(0.040)	(58.410)
DV mean, control	4.26	0.44	439.59
Observations	1990	2052	2038
Panel C. High-SES Households			
	Hours parent helped	Private tutoring	Money on tutoring
Edtech info	0.094	-0.047	40.654
	(0.516)	(0.039)	(157.243)
Data	0.660	0.128**	69.237
	(0.853)	(0.054)	(201.614)
Teacher support	0.512	0.004	121.585
	(0.544)	(0.041)	(150.168)
DV mean, control	4.83	0.52	1013.34
Observations	2191	2242	2213
H vs. L: info	0.688	0.054	0.982
H vs. L: data	0.287	0.117	0.502
H vs. L: teacher	0.724	0.127	0.121

Notes: All expenses reported in taka. Sample includes all Round 2 survey respondents . All regressions include stratification-cell fixed effects and indicators for general information, general and app information, and treatment interactions.. Baseline covariates selected using post-double-selection lasso with 5-fold cross-validation. Robust standard errors are shown in parentheses and clustered at the household level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.22: Impact of outreach on tech-based resource use, disaggregated

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. All						
	Sangsad TV	Video lessons	Robi platform	Teacher remotely	Remote classes	Tech index
Edtech info	-0.063**	-0.052*	-0.015	-0.000	-0.016	-0.085**
	(0.026)	(0.029)	(0.015)	(0.024)	(0.019)	(0.038)
Data	0.007	0.057	-0.019	0.007	-0.002	0.046
	(0.042)	(0.047)	(0.021)	(0.035)	(0.027)	(0.062)
Teacher support	-0.025	0.003	-0.018	0.025	0.008	-0.013
	(0.028)	(0.030)	(0.012)	(0.025)	(0.019)	(0.036)
DV mean, control	0.21	0.27	0.05	0.12	0.08	0.03
Observations	2696	2696	2696	2696	2696	2696
Panel B. Low-SES Households						
	Sangsad TV	Video lessons	Robi platform	Teacher remotely	Remote classes	Tech index
Edtech info	-0.108***	-0.071*	-0.011	-0.013	0.002	-0.106**
	(0.035)	(0.040)	(0.014)	(0.030)	(0.020)	(0.044)
Data	-0.033	0.035	0.021	-0.008	0.004	0.021
	(0.061)	(0.064)	(0.033)	(0.036)	(0.027)	(0.077)
Teacher support	-0.071*	-0.044	-0.014*	0.020	0.025	-0.028
	(0.037)	(0.038)	(0.008)	(0.030)	(0.021)	(0.041)
DV mean, control	0.19	0.22	0.02	0.07	0.03	-0.11
Observations	1255	1255	1255	1255	1255	1255
Panel C. High-SES Households						
	Sangsad TV	Video lessons	Robi platform	Teacher remotely	Remote classes	Tech index
Edtech info	-0.044	-0.040	-0.018	-0.004	-0.013	-0.086
	(0.040)	(0.043)	(0.026)	(0.037)	(0.032)	(0.062)
Data	0.043	0.070	-0.057*	-0.011	0.001	0.042
	(0.061)	(0.072)	(0.030)	(0.058)	(0.045)	(0.098)
Teacher support	0.002	0.046	-0.018	0.024	-0.005	-0.002
	(0.042)	(0.047)	(0.023)	(0.042)	(0.032)	(0.060)
DV mean, control	0.23	0.32	0.08	0.18	0.12	0.14
Observations	1433	1433	1433	1433	1433	1433
H vs. L: info	0.250	0.426	0.733	0.844	0.855	0.892
H vs. L: data	0.521	0.328	0.061	0.737	0.878	0.627
H vs. L: teacher	0.286	0.080	0.980	0.860	0.355	0.490

Notes: Sample includes all Round 1 survey respondents who also completed the R2 survey and learning assessment. All regressions include stratification-cell fixed effects and indicators for general information, general and app information, and treatment interactions.. Baseline covariates selected using post-double-selection lasso with 5-fold cross-validation. Robust standard errors are shown in parentheses and clustered at the household level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.23: Impact of outreach on non-tech-based resource use, disaggregated

	(1)	(2)	(3)	(4)
Panel A. All				
	Textbooks	Exercise books	Teacher in-person	Non-Tech index
Edtech info	-0.004	0.020	0.006	-0.011
	(0.017)	(0.035)	(0.034)	(0.038)
Data	-0.013	0.070	-0.078	-0.054
	(0.025)	(0.051)	(0.050)	(0.058)
Teacher support	-0.002	-0.017	-0.120***	-0.115***
	(0.017)	(0.033)	(0.034)	(0.036)
DV mean, control	0.94	0.33	0.65	0.03
Observations	2696	2696	2696	2696
Panel B. Low-SES Households				
	Textbooks	Exercise books	Teacher in-person	Non-Tech index
Edtech info	0.015	-0.018	0.044	0.032
	(0.025)	(0.053)	(0.052)	(0.054)
Data	-0.039	0.022	-0.106	-0.166*
	(0.053)	(0.075)	(0.081)	(0.091)
Teacher support	0.016	-0.036	-0.147***	-0.087*
	(0.027)	(0.050)	(0.051)	(0.053)
DV mean, control	0.93	0.32	0.61	-0.03
Observations	1255	1255	1255	1255
Panel C. High-SES Households				
	Textbooks	Exercise books	Teacher in-person	Non-Tech index
Edtech info	-0.022	0.040	-0.041	-0.061
	(0.022)	(0.050)	(0.047)	(0.054)
Data	0.025	0.099	-0.038	0.047
	(0.023)	(0.075)	(0.065)	(0.075)
Teacher support	-0.010	-0.018	-0.105**	-0.142***
	(0.022)	(0.049)	(0.049)	(0.052)
DV mean, control	0.96	0.34	0.68	0.08
Observations	1433	1433	1433	1433
H vs. L: info	0.240	0.646	0.150	0.156
H vs. L: data	0.150	0.456	0.507	0.075
H vs. L: teacher	0.397	0.651	0.344	0.606

Notes: Sample includes all Round 1 survey respondents who also completed the R2 survey and learning assessment. All regressions include stratification-cell fixed effects and indicators for general information, general and app information, and treatment interactions.. Baseline covariates selected using post-double-selection lasso with 5-fold cross-validation. Robust standard errors are shown in parentheses and clustered at the household level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.24: Impact of outreach on parental investment

	(1)	(2)	(3)	(4)
Panel A. All				
	Hours parent helped	Private tutoring	Money on tutoring	Money on other education
Edtech info	-0.334	0.030	194.767	-20.820
	(0.521)	(0.032)	(133.374)	(17.636)
Data	0.122	0.031	133.124	9.087
	(0.896)	(0.040)	(148.140)	(25.023)
Teacher support	0.421	-0.070**	97.475	1.530
	(0.550)	(0.031)	(124.357)	(18.435)
DV mean, control	6.69	0.68	1163.64	146.31
Observations	2556	2686	2525	2402
Panel B. Low-SES Households				
	Hours parent helped	Private tutoring	Money on tutoring	Money on other education
Edtech info	-0.902	0.022	156.310	-38.538**
	(0.693)	(0.053)	(143.248)	(16.944)
Data	-0.216	-0.058	195.319	22.241
	(1.257)	(0.071)	(194.835)	(28.235)
Teacher support	0.834	-0.091*	31.955	-19.931
	(0.714)	(0.048)	(111.468)	(20.505)
DV mean, control	6.03	0.62	636.22	81.43
Observations	1186	1249	1193	1114
Panel C. High-SES Households				
	Hours parent helped	Private tutoring	Money on tutoring	Money on other education
Edtech info	0.231	0.023	229.379	-5.403
	(0.821)	(0.042)	(232.540)	(29.554)
Data	0.938	0.095*	-8.557	-9.086
	(1.308)	(0.052)	(253.150)	(41.649)
Teacher support	0.270	-0.039	118.454	7.820
	(0.866)	(0.044)	(235.692)	(29.806)
DV mean, control	7.23	0.73	1618.84	200.63
Observations	1361	1429	1326	1282
H vs. L: info	0.246	0.971	0.682	0.362
H vs. L: data	0.515	0.081	0.804	0.530
H vs. L: teacher	0.474	0.416	0.475	0.410

Notes: All expenses reported in taka. Sample includes all Round 1 survey respondents who also completed the R2 survey and learning assessment. All regressions include stratification-cell fixed effects and indicators for general information, general and app information, and treatment interactions.. Baseline covariates selected using post-double-selection lasso with 5-fold cross-validation. Robust standard errors are shown in parentheses and clustered at the household level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.25: Impact of interventions on student learning (math), endline IPW-adjusted average treatment effect

	(1)	(2)
Panel A. All		
	Unadjusted score	IRT, 2pl
Edtech Info.	0.143**	0.151**
	(0.062)	(0.059)
Edtech Info. + Data	0.028	0.032
	(0.052)	(0.052)
General Info + Data	-0.064	-0.003
	(0.097)	(0.091)
General Info. + Edtech Info. + Data	0.078	0.063
	(0.051)	(0.051)
General Info + Teacher	0.017	-0.008
	(0.059)	(0.058)
General Info + Data + Teacher	-0.029	-0.019
	(0.103)	(0.100)
DV mean, control	0.01	-0.00
Observations	3410	3410
Panel B. Low-SES Households		
	Unadjusted score	IRT, 2pl
Edtech info	0.060	0.108
	(0.094)	(0.090)
Gen. info + data	-0.304**	-0.240*
	(0.136)	(0.124)
Edtech info + data	0.059	0.073
	(0.076)	(0.075)
Gen. info + Edtech info + data	0.009	-0.014
	(0.077)	(0.080)
Gen. info + teacher	0.023	0.063
	(0.089)	(0.084)
Gen. info + data + teacher	0.231	0.163
	(0.153)	(0.144)
DV mean, control	-0.16	-0.21
Observations	1549	1549
Panel C. High-SES Households		
	Unadjusted score	IRT, 2pl
Edtech info	0.215***	0.202***
	(0.078)	(0.075)
Edtech info + data	0.021	0.027
	(0.072)	(0.071)
Gen. info + data	0.157	0.239**
	(0.106)	(0.097)
Gen. info + Edtech info + data	0.129*	0.116*
	(0.066)	(0.064)
Gen. info + teacher	0.011	-0.084
	(0.075)	(0.081)
Gen. info + data + teacher	-0.115	-0.128
	(0.124)	(0.125)
DV mean, control	0.14	0.17
Observations	1861	1861

Notes: Standardized score includes sum of scores on 4 math questions, normalized to the grade-specific control group. IRT adjusted score shows predicted latent ability from full set of math questions, normalized to control group mean (not grade-specific). Sample includes all Round 2 survey respondents. Treatment indicators for receiving general info or general with app info included but not recorded. All regressions include stratification-cell fixed effects. 166 households with missing baseline income excluded. Propensity scores calculated using set of controls listed in Table 1, plus flags for missing values. Robust standard errors are shown in parentheses and clustered at the household level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.26: Impact of interventions on student learning (math), Behaghel et al. (2015) trimming

	Full specification		Trimmed sample	
	(1)	(2)	(3)	(4)
Panel A. All				
	Unadjusted score	IRT, 2pl	Unadjusted score	IRT, 2pl
Edtech info	0.149**	0.150**	0.163***	0.155**
	(0.059)	(0.058)	(0.062)	(0.061)
Data	-0.075	0.048	0.000	0.000
	(0.087)	(0.083)	(.)	(.)
Teacher support	0.001	-0.016	0.000	0.000
	(0.059)	(0.058)	(.)	(.)
DV mean, control				
Observations	3434	3434	1351	1351
Panel B. Low-SES Households				
	Unadjusted score	IRT, 2pl	Unadjusted score	IRT, 2pl
Edtech info	0.023	0.083	-0.016	0.011
	(0.097)	(0.096)	(0.118)	(0.117)
Data	-0.230*	-0.173	0.000	0.000
	(0.139)	(0.131)	(.)	(.)
Teacher support	0.013	0.042	0.000	0.000
	(0.095)	(0.089)	(.)	(.)
DV mean, control				
Observations	1659	1659	559	559
Panel C. High-SES Households				
	Unadjusted score	IRT, 2pl	Unadjusted score	IRT, 2pl
Edtech info	0.248***	0.223***	0.236***	0.202**
	(0.078)	(0.076)	(0.088)	(0.086)
Data	0.046	0.224**	0.000	0.000
	(0.119)	(0.112)	(.)	(.)
Teacher support	-0.029	-0.090	0.000	0.000
	(0.079)	(0.082)	(.)	(.)
DV mean, control				
Observations	1775	1775	642	642

Notes: Standardized score includes sum of scores on 4 math questions, normalized to the grade-specific control group. IRT adjusted score shows predicted latent ability from full set of math questions, normalized to control group mean (not grade-specific). Columns 1 and 2 uses primary specification. Column 3 and 4 restricts restricts the sample to edtech info and the control group households, trimming based on the number of call attempts to achieve an equal response rate between both groups. All regressions include stratification-cell fixed effects. Robust standard errors are shown in parentheses and clustered at the household level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B For Online Publication: Conceptual framework

This section presents a stylized model of the effects of decreasing specific barriers to education when parental educational investments are key inputs. This framework draws upon the literature related to household production and time allocation theory (Becker, 1965; Becker and Tomes, 1976), and it is similar to the one outlined in Todd and Wolpin (2003). We include parental inputs as key contributors to the human capital production function, separating parental time investment (Houtenville and Conway, 2008) from parental monetary investments. We allow for heterogeneity in parental human capital and resources.

Parents maximize utility derived from the child's long-term human capital, H, and the household's present consumption of other goods, C, subject to a human capital production function, a budget constraint, and a time constraint. Assume the utility function is additively separable, increasing, and concave in human capital and in other consumption goods, so that $U(H, C) = u(H) + v(C)$. Households are heterogeneous in parental human capital, θ.

Households invest in human capital via parental time spent on education, i^t, and money for educational activities, i^m. Parents distribute available total household time, T, between labor market supply and teaching their children, (i^t), which could include direct instruction as well as supervising or helping them with their homework. Parents' opportunity cost of teaching is represented by their labor market wage, $w \cdot \theta$, which is increasing in parental human capital θ. The household budget constraint is $C = w \cdot \theta \cdot (T - i^t) - i^m$, where the price of other consumption goods is the numeraire. Hence, higher-skilled parents receive higher incomes and have a higher opportunity cost of time investment. The effectiveness of time investment in teaching also depends on parental human capital, with $i^{t\prime}(\theta) > 0$, so that high-skilled parents are more effective in helping their children.

B.1 Human capital investment through teaching

Human capital is determined by an education production function that relates parental investments and household wealth. All human capital production occurs through effective teaching hours S, which are a function of the effective units of time investment, $i^t(\theta)$ (internal teaching), and the amount of external teaching hours, P, so that $S = f(i^t(\theta), P)$ and $H = g(S)$. $g(\cdot)$ and $f(\cdot)$ are increasing and concave in their arguments. The cost of outside-household teaching hours is c^p, and we assume there is a single external teaching price. Then, parental monetary investment is $i^m = P \cdot c^p$.[28] Note that we are effectively assuming that student effort and motivation do not interact with any of the potential teaching inputs. Households choose the level of monetary investment—through the amount of external teaching they choose— and time investment that maximize their utility, subject to the budget constraint and (expected) human capital production functions:

$$
\begin{aligned}
\max_{P, i^t} \quad & u(H) + v(C) \qquad (1) \\
\text{s.t.} \\
& H = g(S) \\
& S = f(i^t(\theta), P) \\
& C = w \cdot \theta \cdot (T - i^t) - P \cdot c^p
\end{aligned}
$$

For ease of interpretation, we assume households make their investment decisions in two

[28]We assume universal access to textbooks and exercise books at home, so the outside option for human capital generation is to use them without additional guidance, which generates a baseline human capital of α. Without loss of generality, we assume that these resources alone do not produce any human capital, $\alpha = 0$.

steps: first, they decide the amount of external teaching P, and then they choose their time investment, taking i^m as an additional input. Then, the optimization problem yields the following household decision rules: $i^m = \varphi(c^p, \theta)$ and $i^t = \nu(i^m, \theta)$.The absolute number of hours invested by wealthier parents, as well as their relative weight compared to their monetary investment, will depend on this trade-off as well as whether time and monetary investments are perceived substitutes or complements in the human capital production function (i.e., the sign of $\frac{\partial i^t}{\partial i^m}$).

Reduction in cost of teaching hours. We first consider the total effect of decreasing the cost of external teaching hours c^p on human capital *not* holding other inputs constant:

$$\frac{dH}{dc^p} = \frac{\partial g(S)}{\partial S} \cdot \Bigg[\overbrace{\underbrace{\frac{\partial S}{\partial P} \cdot \frac{\partial P}{\partial c^p}}_{\text{Direct effect}} + \underbrace{\frac{\partial S}{\partial P} \cdot \frac{\partial P}{\partial i^m} \cdot \frac{\partial i^m}{\partial c^p}}_{\substack{\text{Behavioral response}\\ \text{monetary investment}}}}^{\substack{\text{External teaching}\\ \text{impacts}}} + \overbrace{\underbrace{\frac{\partial S}{\partial i^t} \cdot \frac{\partial i^t}{\partial i^m} \cdot \frac{\partial i^m}{\partial c^p}}_{\substack{\text{Behavioral response}\\ \text{time investment}}}}^{\substack{\text{Internal teaching}\\ \text{impacts}}} \Bigg] \tag{2}$$

The first term is the direct effect of decreasing the costs of external teaching on human capital, which is weakly positive. The second and third terms capture the impacts of the parental behavioral responses on human capital through changes in monetary and time investments, and the signs of both terms are ambiguous. The overall sign of monetary investments depends on the elasticity of demand for P with respect to c^p, thus reflecting the net income and substitution effects as a result of the cost reduction. The change in human capital due to parental time investments depends on both $\frac{\partial i^m}{\partial c^p}$ as well as $\frac{\partial i^t}{\partial i^m}$; that is, whether parental time and monetary investments are complements ($\frac{\partial^2 g}{\partial i^t \partial i^m} > 0$) or substitutes ($\frac{\partial^2 g}{\partial i^t \partial i^m} < 0$).

B.2 Human capital investment through educational technology

We now expand the conceptual framework to include educational technology as a second, costly, channel of human capital development. We generally specify the technology pricing schedule of the number of (effective) hours spent on the edtech platform, E, as $c^e = \rho(E)$. The cost c^e may be non-linear in the hours of use and can reflect a purchase price, regular subscription fees, or internet data costs associated with its use. Then, economic investment in the edtech platform is $i^e = h(E, c^E)$.

Human capital can now be produced through two different channels, the teaching channel and the edtech channel, $H = g(S, E)$.[29] Parents are uncertain about the value of the edtech platform because it is new. We characterize this uncertainty by differentiating between the *actual* returns of the learning options, $g(\cdot)$, and their *perceived* returns, $\widetilde{g}(\cdot)$ (Boneva and Rauh, 2018). Given the novelty of the edtech platform, at baseline we assume that its perceived returns are very low compared to its actual returns. Formally,

$$\frac{\partial g(S,E)}{\partial E} > \frac{\partial \widetilde{g}(S,E)}{\partial E} \approx 0 \tag{3}$$

Parents now choose the optimal allocation of their time and monetary investment, with the latter being split between external teaching investment and edtech: $C = w \cdot \theta(T -$

[29]The hours spent on the edtech platform E directly enter the human capital production function instead of contributing to S to differentiate inputs that require teaching support (private tutoring, formal schooling, parental help) with an input that students can independently use.

$i^t) - [i^m + i^e]$. It is important to note that their choice of investment inputs will result in a different combination of three human capital production technologies: internal teaching, external teaching, and edtech.

$$
\begin{aligned}
\underset{P,i^t,E}{max} \quad & u(H) + v(C) \\
\text{s.t.} \quad & \\
H &= \widetilde{g}(S, E) \\
S &= f(i^t(\theta), P) \\
C &= w \cdot \theta \cdot (T - i^t) - P \cdot c^p - h(E, \rho(E))
\end{aligned} \tag{4}
$$

Then, the new household decision rules are: $\widetilde{i^m} = \widetilde{\varphi}(\widetilde{i^e}, c^p, \theta)$, $\widetilde{i^e} = \widetilde{\xi}(\widetilde{i^m}, c^e, \theta)$, and $\widetilde{i^t} = \widetilde{\nu}(\widetilde{i^m}, \widetilde{i^e}, \theta)$. Note that the optimal investments $\widetilde{i^t}$, $\widetilde{i^m}$, and $\widetilde{i^e}$ are optimally chosen based on the perceived returns of the technologies $\widetilde{g}(\cdot)$, not on the actual returns.

Information provision. Information about a new technology can signal the value of this new platform, leading individuals to revise their beliefs about its marginal returns upward, hence increasing $\partial\widetilde{g}/\partial E$ toward $\partial g/\partial E$. However, the information on new technologies can more broadly shift the beliefs of the returns to personalized teaching resources as well, making them also revise (upwards or downwards) the returns to teaching, so that $\partial\widetilde{g}/\partial S \lesseqgtr \partial g/\partial S$. Based on this new signal, households may update their educational investments accordingly, investing in the combination of learning options they can afford that will give the highest perceived returns. This implies that, similar to equation (2), an informational policy may generate behavioral responses on monetary and time investments from parents and a re-optimization of the learning resource portfolio above and beyond direct changes in edtech usage.

Reducing edtech costs. One educational policy aimed at reducing budget constraints to accessing education is to decrease the cost of edtech. The total effect of lowering c^e on human capital is:

$$
\begin{aligned}
\frac{dH}{dc^e} = & \overbrace{\frac{\partial g(S,E)}{\partial E} \cdot \left[\underbrace{\frac{\partial E}{\partial c^e}}_{\text{Direct effect}} + \underbrace{\frac{\partial E}{\partial \widetilde{i^e}} \cdot \frac{\partial \widetilde{i^e}}{\partial c^e}}_{\substack{\text{Edtech} \\ \text{investment response}}} \right]}^{\text{Edtech human capital impacts}} \\
& + \overbrace{\frac{\partial g(S,E)}{\partial S} \cdot \left[\underbrace{\frac{\partial S}{\partial P} \cdot \frac{\partial P}{\partial \widetilde{i^m}} \cdot \frac{\partial \widetilde{i^m}}{\partial \widetilde{i^e}} \cdot \frac{\partial \widetilde{i^e}}{\partial c^e}}_{\substack{\text{Behavioral response} \\ \text{monetary investment}}} + \underbrace{\frac{\partial S}{\partial \widetilde{i^t}} \cdot \frac{\partial \widetilde{i^t}}{\partial \widetilde{i^e}} \cdot \frac{\partial \widetilde{i^e}}{\partial c^e}}_{\substack{\text{Behavioral response} \\ \text{time investment}}} \right]}^{\text{Teaching impacts}}
\end{aligned} \tag{5}
$$

Note that although the human capital impacts will be realized through the *actual* learning technology $g(\cdot)$, households' decisions will be based on their beliefs about the learning technology, $\widetilde{g}(\cdot)$, so an informational intervention will not change the partial derivatives, but will change the optimal investments $\widetilde{i^m}$, $\widetilde{i^t}$ and $\widetilde{i^e}$ through the household decision rules.

C For Online Publication: Additional methodological details

C.1 Item response theory

We measure student learning based on a phone-based assessment with students conducted at endline. Partner teachers assisted in creating a bank of math and Bangla test questions aligned with the grade-specific national curriculum that could be asked orally and answered via multiple choice. Each student completed a grade-specific set of four questions per subject set at their 2020 grade level or lower. Based on their performance on these questions, they were then asked four more questions at a slightly lower or slightly higher grade levels. We repeated questions across questionnaires. For example, a math question deemed as ”grade 7” would be asked for students who were in grade 7 as their ”at grade level” questionnaire, asked to students in grade 8 as ’below one level,” asked to students in grade nine as ”below two levels” and to grade 6 as ”above one level”’

We estimate a two-parameter logistic model separately by subject.

C.2 Distribution of answers

With the exception of grade 8 students, very few students answer all or no question correctly in math. Similarly, very few students answer all questions or no questions correctly in Bangla. Overall, 8.9% of the sample is at an endpoint in math, and 6.5% of the sample is at an endpoint in Bangla.

Table C.27: Distribution of test scores, by grade

	Math		**Bangla**	
	Zero correct	All correct	Zero correct	All correct
Grade 6	1.9%	6.1%	0.6%	0.3%
Grade 7	4.6%	6.0%	3.0%	1.8%
Grade 8	3.6%	14.0%	2.4%	0.0%
Grade 9	4.0%	2.6%	3.8%	5.3%
Grade 10	3.5%	0.0%	1.5%	8.6%
All	3.7%	5.2%	2.5%	4.0%

C.2.1 Math

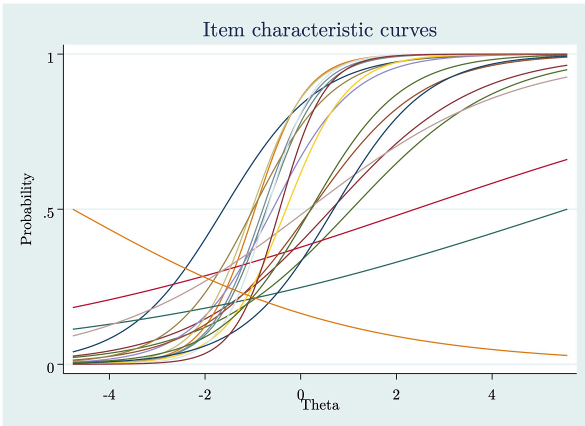
In general, we find that each item has positive discrimination, with well-behaved item characteristic curves:

C.2.2 Bangla

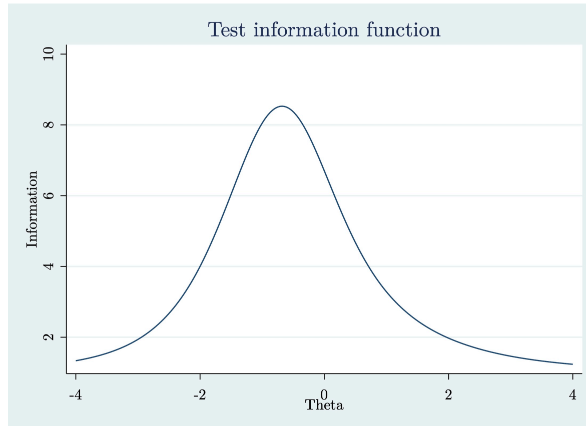
The following curves show that the Bangla results are very noisy. Becuase elements of the curriculum are fully cumulative, we do not expect that a grade 7 would excel at grade 5 questions. We exclude two questions in order to achieve convergence (question 16 and

2-parameter logistic

Item characteristic curve, math

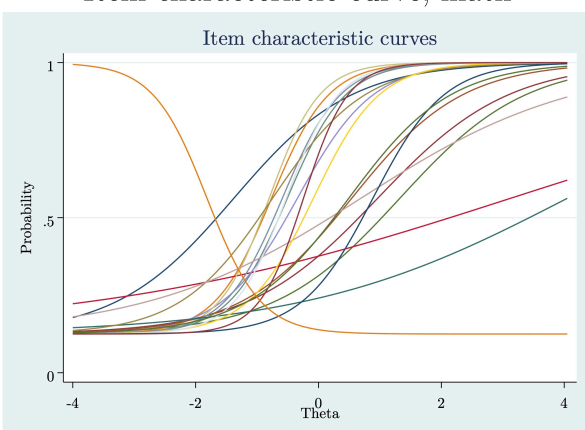


Test information function, math

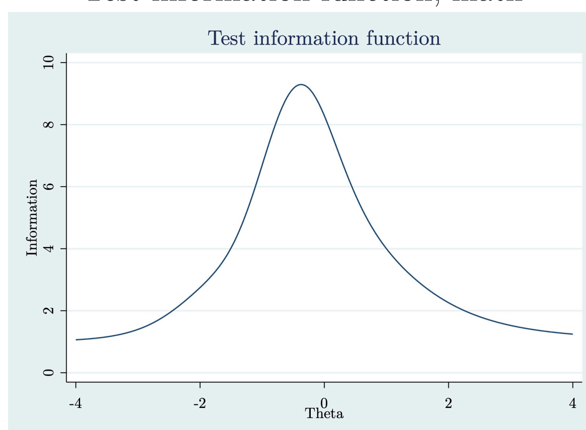


3-parameter logistic

Item characteristic curve, math



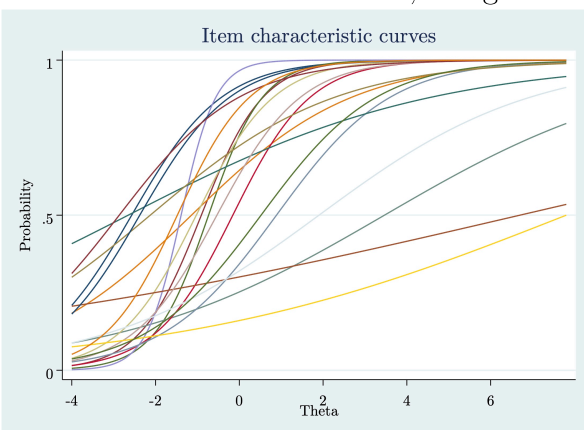
Test information function, math



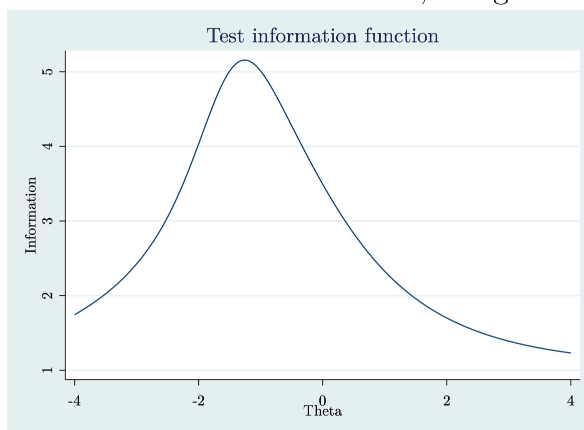
question 76), and we see that the results with the two-parameter model are very different from the three-parameter model results. For these reasons reason, we exclude this subject from our analysis.

2-parameter logisitic

Item characteristic curve, Bangla

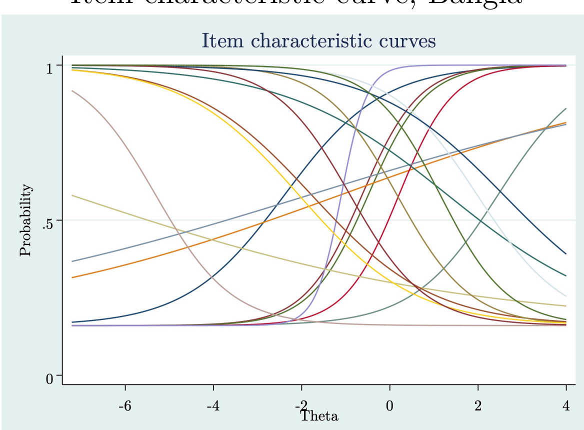


Test information function, Bangla



3-parameter logistic

Item characteristic curve, Bangla



Test information function, Bangla